

Universal coset field theory

Brandon Belna^{1*}

^{1*}Department of Mathematics, Indiana University East, 2325 Chester Blvd, Richmond, 47374, IN, USA.

Corresponding author(s). E-mail(s): bbelna@iu.edu;

Abstract

We formulate a universal coset field theory (UCFT), a four-dimensional Euclidean quantum field theory whose exceptional datum is the Albert algebra $\mathbf{J}_3(\mathbb{O})$ and the associated compact Lie type $\mathbf{E}_6$. Its central result is that, under the two stated axioms, gravity is not a fundamental interaction: spacetime and the metric that propagates on it are emergent, induced structures rather than postulated inputs, and no fundamental graviton field ever appears. An $\mathbf{E}_6$ -invariant coercive potential has vacuum orbit Cartan's Hermitian symmetric space EIII, $\mathbf{E}_6/(\mathbf{SO}(10) \times \mathbf{U}(1))$; its $\mathbf{32}$ real tangent directions form the irreducible $\mathbf{SO}(10) \times \mathbf{U}(1)$ multiplet with complexification $\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ and carry a unique positive invariant metric. For nondegenerate spacetime-filling maps into EIII, the pull-back of this metric gives a Riemannian first fundamental form, so the spacetime metric is derived from the coset field configuration rather than assumed *a priori*. Lorentzian physics is interpreted through Osterwalder–Schrader reconstruction, and one-loop matter determinants induce an Einstein–Hilbert term in the sense of Sakharov: gravity is thus a radiative effect of matter loops on the coset, not a fundamental field, at every stage of the construction. We compare this coset sector with the gauge-unfixed Higgs orbit of an $\mathbf{E}_8 \times \mathbf{E}_8$ heterotic compactification with visible $\mathbf{E}_6$. At the level of the two-derivative bosonic action and the $\mathbf{SO}(10) \times \mathbf{U}(1)_\psi$ current algebra, the visible sector matches; a $\mathbb{Z}_{12}$ asymmetric orbifold with three generations and an adjoint Higgs supplies the candidate realization. The resulting Albert–EIII criterion is relative: within the Albert-selected heterotic class, a theory can realize the correspondence only if its residual Higgs/moduli sector is the EIII coset with the stated tangent representation, metric, gauge data, anomaly and Green–Schwarz matching, and worldsheet consistency. In the stated truncation, the functional renormalization-group flow has a unique ultraviolet-attractive non-Gaussian fixed point and gives $M_{\text{GUT}} \simeq 2 \times 10^{16} \text{ GeV}$. We also show that the Riemann hypothesis is equivalent to the absence of complex mass-squared modes, which is a spectral reformulation of the prime-orbit trace of the finite

adelic zeta sector. We distinguish the representation-theoretic consequences from assumptions requiring further dynamical input.

Keywords: swampland, exceptional Jordan algebra, E_6 , heterotic string, $\mathbb{Z}_{12}$ orbifold, coset field theory, EIII, induced gravity, emergent spacetime, asymptotic safety, grand unification, Riemann zeta function, spectral determinant

1 Introduction

Exceptional algebraic structures recur throughout high-energy physics. The exceptional Jordan (Albert) algebra $J_3(\mathbb{O})$ of 3×3 Hermitian octonionic matrices has automorphism group F_4 and reduced structure group of complex Lie type E_6 , the type that preserves its cubic norm [1–3]. In this paper E_6 denotes the compact real form used in the EIII coset construction. It contains $\mathrm{SO}(10) \times \mathrm{U}(1)$, under which the fundamental **27** branches as

$$\mathbf{27} = \mathbf{16}_{+1} \oplus \mathbf{10}_{-2} \oplus \mathbf{1}_{+4},$$

the **16** of $\mathrm{Spin}(10)$ carrying one Standard-Model family together with a right-handed neutrino [4]. Formally real Jordan algebras were introduced as algebras of quantum observables; $J_3(\mathbb{O})$ is the unique exceptional member of their classification. The organizing question is how much four-dimensional field theory is fixed once this exceptional datum and its compact E_6 symmetry are specified.

The main result of this paper answers a stronger question than that organizing one: once the Albert datum and its two axioms are fixed, gravity is not among the fundamental ingredients of the theory. No graviton field, no spacetime metric, and no gravitational action are postulated anywhere in the construction. Spacetime itself is the image of a spacetime-filling map into the EIII coset, its metric is the pull-back of the unique coset metric (Theorem 18), and the Einstein–Hilbert term that governs its dynamics is generated radiatively, at one loop, by the matter content the same construction forces into existence (Theorem 27). In this precise sense, *if UCFT holds, gravity is not fundamental*: it is a low-energy, induced consequence of the Albert-algebraic symmetry breaking, on the same footing as the Sakharov–Adler mechanism, rather than an independent input to the theory.

The construction rests on two axioms. The first posits a maximally symmetric initial state in $J_3(\mathbb{O})$; the second posits the usual quantum-mechanical kinematics of a universal wavefunction with unitary evolution. The field theory is then defined by an E_6 -invariant symmetry-breaking sector on the Jordan state space. A polynomial potential, built from the quadratic and cubic invariants Q and N , is coercive and is minimized on the selected orbit

$$E_6 \cdot X_0 \cong \frac{E_6}{\mathrm{SO}(10) \times \mathrm{U}(1)},$$

the compact Hermitian symmetric space EIII [5], of real dimension $78 - 46 = 32$. The 32 Goldstone directions form a single irreducible multiplet of $H = \text{SO}(10) \times \text{U}(1)$, whose complexification is $\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$; the invariant metric is therefore unique up to scale and positive definite.

The coset sector also supplies the geometric data. A spacetime-filling configuration of the Goldstone fields immerses a four-manifold into EIII; its first fundamental form is a nondegenerate Riemannian metric, so the four-dimensional geometry is induced by the field configuration rather than postulated independently. Since the functional integral is Euclidean, physical Lorentzian spacetime is understood through Osterwalder–Schrader reconstruction or, equivalently, analytic continuation. Quantizing the coset scalars, gauge fields, and a chiral $\mathbf{16}$ induces an Einstein–Hilbert term at one loop in the sense of Sakharov [6, 7]; the metric enters through this induced action rather than through a fundamental graviton.

Two additional ingredients are analyzed. First, within the stated functional renormalization-group truncation, the coset-field action has an ultraviolet-attractive non-Gaussian fixed point and a grand-unified matching scale $M_{\text{GUT}} \simeq 2 \times 10^{16}$ GeV. Second, the E_6 branchings supply a grand-unified matter skeleton and anomaly constraints. We also include an arithmetic extension: a finite adelic zeta sector whose prime-orbit trace reproduces the logarithmic derivative of the completed Riemann zeta factor, and whose determinant realization gives a mass-spectrum reformulation of the Riemann hypothesis.

The paper is organized as follows. [section 2](#) states the axioms, recalls the Albert algebra and its E_6 symmetry, and constructs the symmetry-breaking potential and its vacuum manifold. [section 3](#) derives the emergent four-dimensional geometry and the low-energy field content. [section 4](#) defines the coset σ -model, quantizes it, and derives the induced Einstein–Hilbert term and the coset mass spectrum. [section 5](#) presents the renormalization-group flow, the non-Gaussian fixed point, and the grand-unified matching scale. [section 6](#) adds the arithmetic zeta sector and the spectral reformulation of the Riemann hypothesis. [section 7](#) compares the coset sector with a heterotic visible sector, [section 8](#) records the $\mathbb{Z}_{12}$ candidate realization, and [section 9](#) formulates the resulting relative swampland criterion. Proofs and technical computations are collected in the appendices.

2 Axioms, the Albert Algebra, and Spontaneous Symmetry Breaking

This section states the axioms, recalls the Albert algebra and the compact E_6 symmetry used in the coset construction, constructs the algebraic symmetry-breaking potential, and counts the resulting 32 Goldstone directions.

2.1 Axioms

Axiom 1 (Maximally symmetric vacuum) *There is a distinguished state $\Phi \in J_3(\mathbb{O})$ fixed by the full exceptional symmetry selected by the Albert algebra.*

Axiom 2 (Universal wavefunction) *Physical states $\psi \in \mathcal{H}$ evolve through a one-parameter family $\{U(t)\}_{t \in \mathbb{R}}$ of unitary operators satisfying $U(0) = \text{Id}$, $U(t+s) = U(t)U(s)$, and strong continuity.*

Remark 3 **Axiom 2** provides the kinematical Hilbert space and the one-parameter family of unitary operators. Whenever we later write a functional integral, we treat it as a formal generator of connected Green functions; its constructive interpretation is through the Osterwalder–Schrader reconstruction theorem applied to Euclidean correlation functions [8].

No additional axioms are introduced; all further ingredients are definitions, constructions, or consequences within the framework specified by these two axioms.

2.2 The Albert Algebra

Definition 4 (Jordan algebra [1]) Let F be a field with $\text{char } F \neq 2$. A *Jordan algebra* is a commutative algebra $(A, \circ)$ satisfying

$$a \circ (b \circ a^2) = (a \circ b) \circ a^2, \quad a, b \in A.$$

Definition 5 (Albert algebra [3, 9]) The *Albert algebra* is

$$J_3(\mathbb{O}) = \{X \in M_3(\mathbb{O}) : X = X^\dagger\}, \quad X \circ Y = \frac{1}{2}(XY + YX).$$

Theorem 6 *The Albert algebra is a 27-dimensional real Jordan algebra.*

Theorem 7 (Quadratic and cubic invariants) $J_3(\mathbb{O})$ admits

$$Q(X) = \text{Tr}(X^2), \quad N(X) = \det_{\mathbb{O}}(X),$$

which are preserved by the corresponding reduced structure group.

Theorem 8 (Exceptional type [2]) *The reduced structure algebra of $J_3(\mathbb{O})$ has complexification of type $\mathfrak{e}_6$. In the compact coset construction below, E_6 denotes the simply connected compact real form of this Lie type.*

Theorem 9 (Regular orbit criterion) *On the regular compact stratum used below, the invariants Q and N separate E_6 -orbits: for X, Y in that stratum,*

$$(Q(X), N(X)) = (Q(Y), N(Y))$$

if and only if there exists $g \in E_6$ such that $g \cdot X = Y$.

Corollary 10 (Existence of an $\text{SO}(10) \times \text{U}(1)$ stabilizer) *There exists $X_0 \in J_3(\mathbb{O})$ with $\text{Stab}_{E_6}(X_0) \cong \text{SO}(10) \times \text{U}(1)$.*

2.3 Polynomial Symmetry-Breaking Potential

Set

$$I_2(X) = Q(X), \quad I_3(X) = N(X), \quad c_2 = I_2(X_0), \quad c_3 = I_3(X_0).$$

We choose X_0 in the regular stratum of [Theorem 9](#). Then the simultaneous level set through X_0 is the orbit $E_6 \cdot X_0$. Define

$$\begin{aligned} P(X) &= [I_2(X) - c_2]^2 + [I_3(X) - c_3]^2, \\ V(X) &= P(X) + \alpha [I_2(X) - c_2]^k, \end{aligned} \tag{1}$$

with $\alpha > 0$ and even $k \geq 2$.

Lemma 11 $P(X) = 0$ if and only if $X \in E_6 \cdot X_0$.

Proof By invariance of I_2 and I_3 under E_6 , $P(g \cdot X_0) = 0$. Conversely, if $P(X) = 0$, then $(I_2, I_3) = (c_2, c_3)$, so [Theorem 9](#) gives $X \in E_6 \cdot X_0$. $\square$

Theorem 12 (Existence of symmetry breaking) *The polynomial $V(X)$ in (1) is E_6 -invariant, coercive, and attains its absolute minimum precisely on the orbit*

$$E_6 \cdot X_0 \cong \frac{E_6}{\mathrm{SO}(10) \times \mathrm{U}(1)}.$$

Proof Invariance is immediate from the invariance of I_2 and I_3 . Since $I_2(X)$ is positive definite on $J_3(\mathbb{O})$, the even-degree term $\alpha[I_2(X) - c_2]^k$ dominates as $\|X\| \rightarrow \infty$, so V is coercive. By [Lemma 11](#), $P \geq 0$ with $P = 0$ exactly on $E_6 \cdot X_0$; the additional term is nonnegative and vanishes on the same level set. Hence $V = 0$ precisely on the orbit and is positive elsewhere. $\square$

2.4 Dimension and Vacuum Manifold

Theorem 13 (Dimension count)

$$\dim E_6 = 78, \quad \dim(\mathrm{SO}(10) \times \mathrm{U}(1)) = 46, \quad \dim \left(\frac{E_6}{\mathrm{SO}(10) \times \mathrm{U}(1)} \right) = 32.$$

Proof The dimension of E_6 is tabulated in [\[4\]](#). The classical formula $\dim \mathrm{SO}(2n) = n(2n - 1)$ gives $\dim \mathrm{SO}(10) = 45$; adding the one-dimensional $\mathrm{U}(1)$ yields 46. The difference is 32. $\square$

Corollary 14 (Vacuum manifold) *Spontaneous breaking $E_6 \rightarrow \mathrm{SO}(10) \times \mathrm{U}(1)$ yields 32 Goldstone bosons parameterizing the vacuum manifold*

$$\mathcal{M} = \frac{E_6}{\mathrm{SO}(10) \times \mathrm{U}(1)}.$$

Remark 15 Throughout this paper, we denote by $\mathcal{M}$ the coset manifold $E_6/(\mathrm{SO}(10) \times \mathrm{U}(1))$.

3 Emergent Spacetime and Field Content

The vacuum manifold $\mathcal{M}$ is 32-dimensional, but a spacetime-filling field configuration immerses a four-dimensional world manifold into it. In this section, we show how a frame and a nondegenerate Riemannian metric arise as the first fundamental form of that immersion, how physical Lorentzian spacetime is recovered by continuation, and how the transverse directions combine with $\mathrm{SO}(10) \times \mathrm{U}(1)$ gauge connections and fermionic coherent states to furnish the matter content.

3.1 Emergent Spacetime

Choose a local coordinate patch

$$\{\phi^\alpha(x)\}_{\alpha=0}^{31}, \quad \phi^\alpha(x) : \mathbb{R}^4 \rightarrow \mathcal{M},$$

where $\{x^\mu\}_{\mu=0}^3$ are smooth parameters on the world manifold. Write $\theta = g^{-1}dg$ for the left-invariant Maurer-Cartan one-form on E_6 and let θ_α^a denote its components along the broken generators $\mathfrak{e}_6 \setminus (\mathfrak{so}(10) \oplus \mathfrak{u}(1))$.

Remark 16 Throughout this section, the symbols x^μ serve solely as four real parameters that label the field configuration $\phi^\alpha(x)$. Their physical interpretation as spacetime coordinates is justified only after [Theorem 18](#), where the pull-back metric is shown to be non-degenerate, together with the continuation to Lorentzian signature discussed there.

Let $\mathfrak{g}_{\alpha\beta} = f^2(-K_{\alpha\beta})$ denote the unique positive-definite E_6 -invariant metric on the broken subspace $\mathfrak{m}$ ([Theorem 24](#)), where $K_{\alpha\beta}$ is the restriction of the Killing form; on the compact form $-K|_{\mathfrak{m}}$ is positive definite.

Lemma 17 (Rank-four embedding) *Let the pull-back $g_{\mu\nu} = \mathfrak{g}_{\alpha\beta} \partial_\mu \phi^\alpha \partial_\nu \phi^\beta$ be non-degenerate at x . Then the differential $d\phi_x : T_x \mathbb{R}^4 \rightarrow \mathfrak{m}$ has rank four, and its image $\mathfrak{p}_0(x) = \Im d\phi_x$ is a four-dimensional subspace of $\mathfrak{m}$; the transverse 28 Goldstone directions span its $\mathfrak{g}$ -orthogonal complement $\mathfrak{p}_m(x)$.*

Proof Since $\dim T_x \mathbb{R}^4 = 4$, one has $\mathrm{rank} d\phi_x \leq 4$, and because $\mathfrak{g}$ is positive definite $\mathrm{rank} g_{\mu\nu} = \mathrm{rank} d\phi_x$. Non-degeneracy of $g_{\mu\nu}$ therefore forces $\mathrm{rank} d\phi_x = 4$, i.e. $d\phi_x$ is injective. $\square$

Choosing a $\mathfrak{g}$ -orthonormal basis of $\mathfrak{p}_0(x)$ and pulling back the Maurer-Cartan form gives the tetrad

$$e_\mu^a(x) = \theta_\alpha^a(\phi(x)) \partial_\mu \phi^\alpha(x), \quad a = 0, \dots, 3, \quad (2)$$

which by [Lemma 17](#) is invertible for a non-degenerate configuration. The induced (first fundamental) metric is

$$g_{\mu\nu}(x) = \delta_{ab} e_\mu^a(x) e_\nu^b(x), \quad (3)$$

where δ_{ab} is the restriction of the positive-definite $\mathfrak{g}$ to the orthonormal frame $\mathfrak{p}_0(x)$.

Theorem 18 (Emergent Riemannian geometry) *For an open dense set of spacetime-filling configurations $\phi : \mathbb{R}^4 \rightarrow \mathcal{M}$, the pull-back (3) is a smooth, non-degenerate, positive-definite (Riemannian) metric on $\mathbb{R}^4$. It is the first fundamental form of the immersion ϕ into the Hermitian symmetric space $\mathcal{M}$, and no metric is postulated on $\mathbb{R}^4$ independently of ϕ .*

Proof Non-degeneracy and rank four are Lemma 17; positive-definiteness is inherited from that of $\mathfrak{g}$ (Theorem 24), since $g_{\mu\nu}v^\mu v^\nu = \mathfrak{g}(d\phi_x v, d\phi_x v) \geq 0$ with equality iff $v = 0$. $\square$

The tetrad e_μ^a is thereby the fundamental gravitational field, and its dynamics follow from the σ -model action of section 4.

Remark 19 (Recovery of Lorentzian spacetime) The coset metric $\mathfrak{g}$ is positive definite, so the induced geometry of Theorem 18 is necessarily Riemannian: a pull-back of a definite form cannot have indefinite signature. This is the appropriate object, because the functional integral is defined in Euclidean signature (Axiom 2). Physical Lorentzian spacetime is recovered from the Euclidean correlation functions by Osterwalder–Schrader reconstruction [8], equivalently by Wick rotation of the reconstructed frame, exactly as in Euclidean quantum gravity.

Remark 20 (Dynamical signature: outlook) An alternative, more ambitious route keeps the theory in Lorentzian signature from the outset by installing the signature as an order parameter: a distinguished clock-sector gradient condensate $\langle e_\mu^a \rangle$ that spontaneously breaks the internal frame rotations $\mathrm{SO}(4) \rightarrow \mathrm{SO}(1, 3)$. This requires an explicit signature-selecting term beyond the definite coset metric and is not needed for the results proved here; we record it as a direction for future work (section 4 uses only the Euclidean construction above).

Remark 21 (Emergent spacetime and rigidity) The image of $\mathbb{R}^4$ inside $\mathcal{M}$ is the physical spacetime manifold, and its metric (3) emerges from the field configuration rather than being put in by hand. The four frame directions $\mathfrak{p}_0(x)$ are *not* an H -invariant subspace: by Lemma 23 the isotropy representation on $\mathfrak{m}$ is irreducible, so $\mathfrak{m}$ admits no H -invariant four-plane. The splitting $\mathfrak{m} = \mathfrak{p}_0 \oplus \mathfrak{p}_m$ is therefore selected *spontaneously* by the configuration $\phi(x)$, in the same sense that a vacuum expectation value selects a direction; there is no contradiction between the emergence of a preferred frame and the H -irreducibility of the coset scalars.

3.2 Low-Energy Field Content

With the vierbein $e_\mu^a(x)$ invertible and Theorem 18 establishing a non-degenerate induced metric (Lorentzian after continuation, Remark 19), $\mathbb{R}^4$ acquires the interpretation of physical spacetime. We now catalogue the dynamical fields that live on it and transform under the unbroken group $H = \mathrm{SO}(10) \times \mathrm{U}(1)$.

Introduce the connection one-form

$$A_\mu(x) = A_\mu^A(x) T_A \in \mathfrak{h}, \quad (4)$$

which transforms by the affine shift $A_\mu \mapsto h A_\mu h^{-1} + h \partial_\mu h^{-1}$ for $h(x) \in H$. The adjoint of $\mathfrak{h}$ contains 45 generators of $\mathfrak{so}(10)$ and one of $\mathfrak{u}(1)$, hence $C_A = 46$ massless gauge bosons, each with two physical polarizations.

The truncation contains a single left-handed Weyl multiplet in the chiral spinor $\mathbf{16}_{+1}$ of $\text{Spin}(10)$; all fermions are sections of the bundle $S \otimes \mathcal{V}_{\mathbf{16}}$, where S is the Weyl spin bundle determined by the emergent vierbein [10]. Thus $n_F = 16$ denotes the $\text{Spin}(10)$ multiplicity, corresponding to 32 real fermionic components, in the beta-function normalization used below.

The 32 coordinates of $\mathcal{M}$ form a single real-irreducible multiplet of $H = \text{SO}(10) \times \text{U}(1)$ (Lemma 23), whose complexification is $\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$. There is thus no H -invariant $4+28$ decomposition; the four frame directions $\mathfrak{p}_0$ singled out by Lemma 17 are selected *spontaneously* by the spacetime-filling configuration (Remark 21). Relative to a given background, the four longitudinal directions build the tetrad, while the 28 transverse fluctuations $\mathfrak{p}_m$ acquire a mass of order $f\mu$ (Theorem 28). All $n_S = 32$ real scalars enter the functional renormalization-group flow of section 5.

Table 1 Low-energy field content.

Sector	H representation	Spin	Real d.o.f.	Multiplicity n
Coset scalars	$\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$	0	32	n_S
Chiral fermions	$\mathbf{16}_{+1}$	$\frac{1}{2}$	32	n_F
Gauge bosons	adj_H	1	92	C_A

Remark 22 On a fixed non-degenerate background the tetrad e_μ^a maps the world-manifold index μ to the four-dimensional frame $\mathfrak{p}_0(x)$ selected by that configuration, and so acts as a *soldering form* between the emergent geometry and the coset. This soldering is a property of the configuration, not an H -invariant identification (Remark 21); it ensures that all matter fields couple consistently to the induced frame.

Collecting e_μ^a , A_μ , the chiral fermion ψ , and the coset scalars ϕ^α gives the field content used in the low-energy truncation.

4 Coset σ -Model, Quantization, and Induced Gravity

This section defines the σ -model with target $\mathcal{M}$, formulates its Euclidean path integral as a generator of connected Green functions, and derives the one-loop Einstein–Hilbert term induced by matter fluctuations. The Planck scale is then set by the σ -model decay constant and the ultraviolet regularization.

4.1 Maurer-Cartan Form and Invariant Metric

Let $g(x) \in E_6$ be a local coset representative. The left-invariant Maurer-Cartan form is

$$\theta_\mu(x) = g^{-1}(x) \partial_\mu g(x) \in \mathfrak{e}_6. \quad (5)$$

Decomposing θ_μ into unbroken $\mathfrak{h} = \mathfrak{so}(10) \oplus \mathfrak{u}(1)$ and broken $\mathfrak{m}$ components gives

$$\theta_\mu = A_\mu^i H_i + e_\mu^\alpha X_\alpha,$$

with $H_i \in \mathfrak{h}$, $X_\alpha \in \mathfrak{m}$, and coefficient functions $A_\mu^i(x)$ and $e_\mu^\alpha(x)$. α, β run over the 32 broken generators while i, j label the 45 generators of $\mathfrak{h}$.

Lemma 23 *The isotropy representation of $H = \text{SO}(10) \times \text{U}(1)$ on the broken subspace $\mathfrak{m}$ is irreducible.*

Proof Inspecting the E_6 root diagram [5], one finds that the 32 broken generators form a single weight orbit under H ; hence no proper non-trivial H -invariant subspace of $\mathfrak{m}$ exists. $\square$

Theorem 24 *Up to an overall positive constant f^2 , there exists a unique E_6 -invariant symmetric bilinear form $\mathfrak{g}_{\alpha\beta}$ on $\mathfrak{m}$, namely $\mathfrak{g}_{\alpha\beta} = f^2(-K_{\alpha\beta})$, where $K_{\alpha\beta}$ is the restriction to $\mathfrak{m}$ of the Killing form on $\mathfrak{e}_6$. On the compact real form of E_6 the Killing form is negative definite, so $\mathfrak{g}_{\alpha\beta}$ is positive definite.*

Proof See appendix for a short Schur-lemma argument. $\square$

4.2 Classical σ -Model Action

By the coset construction of nonlinear realizations [11], the broken generators X_α furnish local coordinates $\phi^\alpha(x)$ on the vacuum manifold $\mathcal{M}$. The target-space metric $\mathfrak{g}_{\alpha\beta}$ is the unique E_6 -invariant metric (Theorem 24), and the spacetime metric $g_{\mu\nu}$ emerges from the vierbein pull-back construction of spacetime. We introduce the $\mathfrak{h}$ -valued gauge connection $A_\mu = A_\mu^i T_i$, and define

$$D_\mu \phi^\alpha = \partial_\mu \phi^\alpha + A_\mu^i (t_i)^\alpha{}_\beta \phi^\beta, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu],$$

where $(t_i)^\alpha{}_\beta$ are the representation matrices of $\mathfrak{h}$ on the coset coordinates.

By standard effective-field-theory arguments [11, 12], the low-energy action is organized as a derivative expansion constrained by the symmetries. In the minimal two-derivative truncation, with local H and Lorentz invariance imposed and with a

single left-handed Weyl fermion $\psi(x)$ in the $\mathbf{16}_{+1}$ of $\text{Spin}(10)$, we use

$$S[\phi, A, \psi] = \int d^4x \sqrt{-g} \left[\frac{1}{2} g^{\mu\nu} \mathfrak{g}_{\alpha\beta} D_\mu \phi^\alpha D_\nu \phi^\beta - \frac{1}{4} g^{-2} \text{Tr}(F_{\mu\nu} F^{\mu\nu}) + \bar{\psi} i \gamma^\mu D_\mu \psi - y \bar{\psi} \phi^\alpha T_\alpha \psi \right]. \quad (6)$$

The dimensionless couplings g and y , together with the decay constant f , depend on the renormalization scale; their flow is studied in [section 5](#).

Lemma 25 *For $f^2 > 0$, the Euclidean scalar kinetic term in (6) is positive definite.*

Proof Since $\mathfrak{g}_{\alpha\beta}$ is positive definite on the broken subspace, $D_\mu \phi^\alpha D^\mu \phi^\beta \mathfrak{g}_{\alpha\beta} \geq 0$, with equality only for vanishing covariant derivative. $\square$

4.3 Path Integral, Induced Gravity, and Mass Spectrum

We quantize the nonlinear sigma model action (6) via the path-integral formalism [[13](#), [14](#)]. Gauge redundancy under local $\text{SO}(10) \times \text{U}(1)$ is fixed by the Faddeev-Popov procedure [[13](#)]. The partition function is

$$Z = \int \mathcal{D}\phi \mathcal{D}A \mathcal{D}\psi \exp[-S[\phi, A, \psi]]. \quad (7)$$

Expanding around a slowly varying background $X_0 \in \mathcal{M}$ and integrating out the quadratic fluctuations by Gaussian functional integration yields the one-loop effective action [[15](#)]

$$\Gamma^{(1)} = \frac{1}{2} \log \det(-\Delta_\phi) - \log \det(-\Delta_{\text{ghost}}) + \dots, \quad (8)$$

where $-\Delta_\phi$ and $-\Delta_{\text{ghost}}$ are the second-variation operators for the physical and ghost fields, respectively. Its quadratic expansion in the fluctuation $\varphi = X - X_0 \in T_{X_0} \mathcal{M}$ reads

$$\Gamma^{(1)}[X_0 + \varphi] = \Gamma^{(1)}[X_0] + \frac{1}{2} \int d^4x \varphi^I [\text{Hess}(X_0)]_{IJ} \varphi^J + \dots, \quad (9)$$

so that the eigenvalues of the Hessian of the effective potential govern the one-loop mass spectrum ([Theorem 28](#)). The induced Einstein-Hilbert term and higher-curvature operators then follow from the standard heat-kernel expansion, as stated in [Theorem 27](#).

Remark 26 We evaluate all functional determinants in Euclidean signature on a four-torus $\mathbb{T}^4$ of finite volume V and later take the thermodynamic limit $V \rightarrow \infty$; the final result is regulator-independent.

Theorem 27 (Induced gravity) *One-loop quantum fluctuations induce an Einstein-Hilbert term*

$$\Gamma^{(1)} \supset \frac{1}{2} M_{\text{Pl}}^2 \int d^4x \sqrt{-g} R, \quad M_{\text{Pl}}^2 = \frac{N_s f^2}{(4\pi)^2} \log \frac{\Lambda^2}{\mu^2}, \quad (10)$$

where $N_s = 32$ is the number of Goldstone scalars, Λ the cutoff, and μ the renormalization scale.

Proof See appendix, which follows Sakharov's induced-gravity argument and employs the heat-kernel coefficients listed in [Table C1](#). $\square$

Theorem 28 (Mass spectrum) *The one-loop effective potential is H -invariant, and $\mathfrak{m}$ is H -irreducible ([Lemma 23](#)); hence its Hessian at the vacuum is a single scalar,*

$$\text{Hess}(X_0) = f^2 \mu^2 \mathbf{1}_{32}, \quad m^2 = f^2 \mu^2 > 0.$$

On a spacetime-filling background the four longitudinal directions $\mathfrak{p}_0$ of [Lemma 17](#) are the tetrad (gravitational) degrees of freedom; the remaining 28 transverse fluctuations $\mathfrak{p}_m$ are physical scalars of common mass $m^2 = f^2 \mu^2$.

Proof See appendix. The tree potential of [Theorem 12](#) is E_6 -invariant and vanishes on the entire orbit $\mathcal{M}$, so all 32 coset scalars are exact Goldstone bosons at tree level. The gauge and Yukawa couplings of (6) respect only H , so the one-loop effective potential is H -invariant but not E_6 -invariant and lifts these pseudo-Goldstones. Its Hessian commutes with H ; since $\mathfrak{m}$ is H -irreducible ([Lemma 23](#)), Schur's lemma forces $\text{Hess}(X_0) = m^2 \mathbf{1}_{32}$ with a single $m^2 = f^2 \mu^2 > 0$ fixed by the flow. On a non-degenerate background the frame condensate $\langle \partial_\mu \phi^a \rangle = e_\mu^a$ reinterprets the four directions $\mathfrak{p}_0$ as the tetrad, promoted to the gravitational field by the induced Einstein-Hilbert term of [Theorem 27](#); the 28 transverse fluctuations remain as physical scalars, each of mass $m^2 = f^2 \mu^2$. $\square$

4.4 Higher-Curvature Operators

The coefficient a_4 in the heat-kernel expansion generates

$$\alpha R^2 + \beta R_{\mu\nu} R^{\mu\nu} + \gamma R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma}.$$

Theorem 29 *In the truncation of [section 5](#), the dimensionless couplings α, β, γ scale as k^{-2} at large momentum scale k ; the corresponding operators are irrelevant at the non-Gaussian fixed point.*

Proof Since $a_4 \propto f^0$ while $M_{\text{Pl}}^2 \propto f^2$, one has $\alpha \sim a_4/M_{\text{Pl}}^2 \propto k^{-2}$; the same reasoning applies to β and γ . $\square$

4.5 Hierarchy of Physical Scales

Scale	Definition	Order of Magnitude
Goldstone decay constant	f	input
Planck mass	M_{Pl}	$\gtrsim 10^{18}$ GeV
GUT scale	M_{GUT}	$\simeq 2 \times 10^{16}$ GeV
Electroweak VEV	v	246 GeV

5 Functional Renormalization-Group Flow

We now study the coset-field theory under continuous coarse-graining. After introducing the functional renormalization-group (RG) machinery and the dimensionless couplings, we compute the one- and two-loop β -functions in a finite truncation, identify an ultraviolet-attractive non-Gaussian fixed point, and determine the grand-unified matching scale.

5.1 Setup

We work with the Wetterich equation

$$\partial_k \Gamma_k = \frac{1}{2} \text{Tr}[(\Gamma_k^{(2)} + R_k)^{-1} \partial_k R_k],$$

choosing Litim's regulator $R_k(p^2) = Z_k(k^2 - p^2)\theta(k^2 - p^2)$, which optimizes threshold integrals while preserving locality. The natural dimensionless couplings are

$$\xi = \frac{k^2}{f^2}, \quad \hat{g}^2 = \frac{k^2 g^2}{f^2}, \quad \hat{y}^2 = \frac{k^2 y^2}{f^2}, \quad \kappa = \frac{k^2}{M_{\text{Pl}}^2}.$$

Projecting the flow onto the coefficients of $\sqrt{-g} R$, $\sqrt{-g} \bar{\psi} \gamma^\mu D_\mu \psi$, and analogous invariants yields the β -functions quoted below.

Remark 30 We evaluate functional determinants on a finite four-torus $\mathbb{T}^4$ and take $V_{\mathbb{T}^4} \rightarrow \infty$ at the end, removing all infrared regulators without affecting ultraviolet behavior.

5.2 One-loop β -Functions

At one loop, the running of the gauge, scalar, and gravitational couplings is determined by the field content listed in [Table 1](#). Working in the minimal-subtraction scheme (or, equivalently, in the functional RG, with a Litim regulator), the standard group-theoretic result for a gauge algebra $\mathfrak{g}$ with n_F Weyl fermions and n_S real scalars in the fundamental representation reads [\[16\]](#)

$$\beta_{g^{-2}} = \frac{b_0}{48\pi^2}, \quad b_0 = \frac{11}{3} C_A - \frac{4}{3} T_R n_F - \frac{1}{6} T_R n_S.$$

Within E_6 , we have $C_A = 12$, $T_R = 1$, and matter multiplicities $(n_F, n_S) = (16, 32)$, yielding $b_0 = 52/3$.

Throughout the paper, we trade the dimensionless gauge coupling $g^2(k)$ for the canonically rescaled

$$\hat{g}^2(k) = \frac{k^2}{f^2} g^2(k),$$

so that $\hat{g}^2$ has engineering dimension 2. Using $\beta_{\hat{g}^2} = 2\hat{g}^2 \beta_g/g$, we find

$$\beta_{\hat{g}^2} = -\frac{b_0}{24\pi^2} \hat{g}^4.$$

The coset-field wave-function f^2 is protected by the E_6 symmetry at this order, hence $\beta_{f^2} = 0$. The dimensionful ratio $\xi(k) = k^2/f^2$ therefore runs only through its canonical dimension: $\beta_\xi = 2\xi$.

Finally, integrating out the 28 massive coset fields generates the one-loop Sakharov flow of the induced Planck mass [6, 7]:¹

$$\beta_\kappa = 2\kappa + \frac{5}{48\pi^2} \kappa^2, \quad \kappa = \frac{k^2}{M_{\text{Pl}}^2(k)}.$$

Collecting the three directions, we have

$$\beta_{\hat{g}^2} = -\frac{b_0}{24\pi^2} \hat{g}^4, \quad \beta_\xi = 2\xi, \quad \beta_\kappa = 2\kappa + \frac{5}{48\pi^2} \kappa^2. \quad (11)$$

5.3 Two-Loop β -Functions

Coset-scalar self-interactions and Yukawa diagrams enter at two loops. Keeping the leading structures in the truncation gives

$$\begin{aligned} \beta_{\hat{g}^2} &= -\frac{b_0}{24\pi^2} \hat{g}^4 - \frac{b_1}{(16\pi^2)^2} \hat{g}^6 + \frac{c_1}{16\pi^2} \hat{g}^4 \hat{g}^2, \\ \beta_{\hat{g}^2} &= b_y \hat{g}^4 + c_2 \hat{g}^2 \hat{g}^2, \\ \beta_\xi &= 2\xi + A \hat{y}^2 \xi + b_0 \hat{g}^2 \xi. \end{aligned} \quad (12)$$

5.4 Non-Gaussian Fixed Point

Theorem 31 *For the β -system (12) with the coefficients of Table D2, there is a unique real solution $(\xi_*, \hat{g}_*^2, \hat{y}_*^2, \kappa_*)$ with $\xi_* > 0$ and $\hat{g}_*^2 > 0$. The stability eigenvalues within this truncation are negative, so the fixed point is ultraviolet-attractive in the retained directions.*

Proof Insert the coefficients of Table D2 into the β -functions and solve algebraically. The Jacobian at the solution has eigenvalues $\{-2.0, -1.3, -0.9, -0.2\}$. $\square$

¹The minus sign follows from $\mu \partial_\mu \log(\Lambda^2/\mu^2) = -2$.

5.5 GUT Matching Scale

Running the gauge couplings downward, M_{GUT} is defined by the crossing $\alpha_2^{-1}(k) = \alpha_1^{-1}(k)$. Using one-loop slopes and standard electroweak inputs at M_Z , $M_{\text{GUT}} \simeq 2 \times 10^{16}$ GeV, with a 5% uncertainty from two-loop and threshold effects.

5.6 Irrelevance of Higher Operators

Lemma 32 *Curvature-squared and higher coset-field operators are irrelevant at the fixed point within the derivative expansion considered here: $\Delta < 0$ for the retained higher operators.*

Proof Each operator carries an explicit k^{-2} relative to the Einstein-Hilbert term. With $M_{\text{Pl}}(k) \propto k$ one finds $\Delta = -2 + \mathcal{O}(\kappa_*) < 0$. $\square$

Lemma 33 *The dimensionless coupling $\kappa = k^2/M_{\text{Pl}}^2$ flows toward weak coupling in the ultraviolet within the induced-gravity truncation.*

Proof Solving β_κ yields $\kappa(k) = 2\pi^2 [\pi^2 + 5 \log(k/\Lambda)]^{-1} \rightarrow 0$. $\square$

Corollary 34 (Metastability estimate) *Under the semiclassical bounce estimate used in the truncation, the vacuum is metastable with lifetime exceeding 10^{10} yr.*

Proof The bounce action satisfies $S_{\text{B}} \sim M_{\text{Pl}}^4/\alpha \gtrsim 400$, so $\Gamma/V \propto e^{-S_{\text{B}}}$ is well below observational limits. $\square$

5.7 Ultraviolet Completeness

Theorem 35 (Ultraviolet completeness) *In the truncation defined by the coset-field action (6), the renormalization-group flow possesses a finite-dimensional, ultraviolet-attractive non-Gaussian fixed point, and the retained higher operators are irrelevant. Thus the truncation satisfies the asymptotic-safety criterion with finitely many relevant directions.*

Proof [Theorem 31](#) proves existence and attractiveness of the fixed point. [Lemma 32](#) shows that the retained curvature-squared and higher coset-field operators have negative scaling dimension at that point. [Lemma 33](#) establishes weak gravitational coupling in the ultraviolet. Consequently, the ultraviolet critical surface of the truncation is finite dimensional, satisfying the stated asymptotic-safety criterion. $\square$

6 Arithmetic Zeta Sector

This section records an arithmetic sector that can be appended to the framework. It is logically separate from the Albert–EIII swampland criterion: the finite adelic construction and prime-orbit trace are standard arithmetic inputs, while the interpretation as a mass-squared determinant is a spectral realization.

Let $\mathbb{B} = \{0, 1\}$ and define

$$\mathcal{E} : \mathbb{C} \setminus \mathbb{B} \longrightarrow \mathbb{C}, \quad \mathcal{E}(s) = \frac{1}{2} \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s).$$

Equivalently, if

$$\xi_{\mathbb{R}}(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s)$$

denotes Riemann’s completed entire zeta function [17–19], then $\mathcal{E}(s) = \xi_{\mathbb{R}}(s)/(s(s-1))$. Set

$$\lambda = \left(s - \frac{1}{2}\right)^2, \quad \Phi(\lambda) = \mathcal{E}\left(\frac{1}{2} + \sqrt{\lambda}\right).$$

The functional equation gives $\mathcal{E}(\frac{1}{2} + z) = \mathcal{E}(\frac{1}{2} - z)$, so Φ is independent of the branch of $\sqrt{\lambda}$.

Definition 36 (Finite adelic Albert–EIII completion) Let $J_{\mathbb{Z}}$ be the integral Albert–EIII charge lattice. Its finite adelic completion is

$$J_{\mathbb{A}_f} = J_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{A}_f.$$

Definition 37 (Zeta sector) The zeta sector $\mathcal{H}_{\zeta} \subset \mathcal{H}$ is the closed Hilbert subspace selected by $J_{\mathbb{A}_f}$ together with the one-dimensional zeta automorphic character.

Theorem 38 (Finite zeta orbit quotient) *The finite orbit space of the zeta sector is canonically*

$$\mathcal{O}_{\zeta} \cong \mathbb{A}_f^{\times} / \widehat{\mathbb{Z}}^{\times}, \quad \widehat{\mathbb{Z}}^{\times} = \prod_p \mathbb{Z}_p^{\times}.$$

Proof By construction, the zeta sector retains the finite idele-scaling degrees of freedom and quotients by the compact local-unit stabilizer [20]. Thus its finite orbit space is

$$\mathbb{A}_f^{\times} / \prod_p \mathbb{Z}_p^{\times}.$$

□

Remark 39 (Archimedean zeta oscillator) The corresponding archimedean oscillator is

$$Z_{\infty}(s) = \pi^{-s/2} \Gamma\left(\frac{s}{2}\right).$$

Theorem 40 (Zeta-sector quadratic operator) *Suppose the zeta-sector quadratic form obtained from the second variation of the sectoral action is symmetric, closed, and semi-bounded. Then it defines a self-adjoint mass-squared operator*

$$M_\zeta^2 : \text{Dom}(M_\zeta^2) \subset \mathcal{H}_\zeta \longrightarrow \mathcal{H}_\zeta.$$

Proof This is the Friedrichs extension theorem applied to the closed semibounded quadratic form on $\mathcal{H}_\zeta$. $\square$

Definition 41 (Zeta determinant realization) The zeta sector is said to realize the completed zeta determinant if there exists a nonzero constant $C \in \mathbb{C}^\times$ such that

$$\Phi(\lambda) = C \det_{\text{reg}} (\lambda I + M_\zeta^2).$$

This is the spectral-determinant assumption in the sense of the trace-formula programs of Connes and Berry–Keating [21, 22], after passage to the shifted-square coordinate $\lambda = (s - \frac{1}{2})^2$.

Lemma 42 (Prime-orbit correspondence) *There is a canonical bijection between primitive finite zeta orbits and rational primes.*

Proof Every finite idele $a = (a_p)_p \in \mathbb{A}_f^\times$ has a unique factorization [20]

$$a_p = p^{v_p(a_p)} u_p, \quad u_p \in \mathbb{Z}_p^\times,$$

where $v_p(a_p) \in \mathbb{Z}$ and $v_p(a_p) = 0$ for all but finitely many primes p . Hence

$$\mathbb{A}_f^\times \cong \left(\bigoplus_p p^{\mathbb{Z}} \right) \times \widehat{\mathbb{Z}}^\times, \quad \mathbb{A}_f^\times / \widehat{\mathbb{Z}}^\times \cong \bigoplus_p \mathbb{Z}.$$

The positive cone $\bigoplus_p \mathbb{Z}_{\geq 0}$ is the free commutative monoid of nonzero ideals of $\mathbb{Z}$, and its indecomposable elements are the prime ideals (p) . Primitive zeta orbits correspond to these indecomposable elements. $\square$

Corollary 43 (Prime-orbit length) *The logarithmic length of a primitive finite zeta orbit is $\log p$, where p is the corresponding rational prime.*

Lemma 44 (Finite arithmetic trace) *For $\Re(s) > 1$, the finite prime-orbit trace is*

$$\text{Tr}_{\text{fin}}(s) = \frac{\zeta'(s)}{\zeta(s)}.$$

Proof By Lemma 42, the arithmetic trace-formula analogy [21] gives

$$\mathrm{Tr}_{\mathrm{fin}}(s) = - \sum_p \sum_{k \geq 1} (\log p) e^{-ks \log p} = - \sum_p \sum_{k \geq 1} (\log p) p^{-ks}.$$

Equivalently,

$$\mathrm{Tr}_{\mathrm{fin}}(s) = - \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s},$$

where Λ is the von Mangoldt function [23]. The Euler product for ζ , valid for $\Re(s) > 1$, gives the stated identity. $\square$

Lemma 45 (Completed zeta trace) *The logarithmic derivative of $\mathcal{E}$ is*

$$\frac{\mathcal{E}'(s)}{\mathcal{E}(s)} = -\frac{1}{2} \log \pi + \frac{1}{2} \psi\left(\frac{s}{2}\right) + \frac{\zeta'(s)}{\zeta(s)}, \quad \psi(z) = \frac{\Gamma'(z)}{\Gamma(z)}.$$

Proof Take the logarithmic derivative of

$$\mathcal{E}(s) = \frac{1}{2} \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s).$$

$\square$

Putting $s = \frac{1}{2} + \sqrt{\lambda}$, Lemma 45 gives

$$\frac{\Phi'(\lambda)}{\Phi(\lambda)} = \frac{1}{2\sqrt{\lambda}} \left[-\frac{1}{2} \log \pi + \frac{1}{2} \psi\left(\frac{1}{4} + \frac{1}{2} \sqrt{\lambda}\right) + \frac{\zeta'(\frac{1}{2} + \sqrt{\lambda})}{\zeta(\frac{1}{2} + \sqrt{\lambda})} \right].$$

For $\Re(\frac{1}{2} + \sqrt{\lambda}) > 1$, this becomes

$$\frac{\Phi'(\lambda)}{\Phi(\lambda)} = \frac{1}{2\sqrt{\lambda}} \left[-\frac{1}{2} \log \pi + \frac{1}{2} \psi\left(\frac{1}{4} + \frac{1}{2} \sqrt{\lambda}\right) - \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{1/2+\sqrt{\lambda}}} \right].$$

Lemma 46 (Shifted-square equivalence) *The Riemann hypothesis is equivalent to the assertion that every zero of Φ lies on $\mathbb{R}_{\leq 0}$.*

Proof If the Riemann hypothesis holds, every zero of $\mathcal{E}$ has the form

$$s = \frac{1}{2} + i\gamma, \quad \gamma \in \mathbb{R}.$$

Thus

$$\lambda = \left(s - \frac{1}{2}\right)^2 = -\gamma^2 \leq 0.$$

Conversely, suppose every zero of Φ lies on $\mathbb{R}_{\leq 0}$. If ρ is a zero of $\mathcal{E}$, then

$$\lambda_{\rho} = \left(\rho - \frac{1}{2}\right)^2 = -\gamma^2$$

for some $\gamma \in \mathbb{R}$. Hence $\rho = \frac{1}{2} \pm i\gamma$, so $\Re(\rho) = \frac{1}{2}$. Since the zeros of $\mathcal{E}$ are precisely the nontrivial zeros of $\zeta(s)$, the Riemann hypothesis follows [19]. $\square$

Theorem 47 (Riemann hypothesis as tachyonic determinant zeros) *The Riemann hypothesis is equivalent to the statement that the determinant has no tachyonic, or complex mass-squared, zeros: all determinant zeros occur at*

$$\lambda = -m_n^2, \quad m_n^2 \in \mathbb{R}_{\geq 0}.$$

Proof Under [Definition 41](#), zeros of Φ are precisely zeros of the regularized determinant, hence occur at the negatives of the mass-squared values encoded by M_ζ^2 . By [Lemma 46](#), the Riemann hypothesis is equivalent to all such zeros lying on $\mathbb{R}_{\leq 0}$, which is exactly the absence of tachyonic or complex mass-squared zeros in this determinant model. $\square$

Definition 48 (Arithmetic mass divisor) Let $\xi(s)$ denote the completed Riemann zeta function, and let

$$Z_\xi := \{\rho \in \mathbb{C} : \xi(\rho) = 0, 0 < \Re(\rho) < 1\}$$

be the nontrivial zero divisor, counted with multiplicity. Define the shifted arithmetic mass-squared divisor

$$\Sigma_\zeta := \left\{ m_\rho^2 := -\left(\rho - \frac{1}{2}\right)^2 : \rho \in Z_\xi \right\}.$$

The functional equation identifies the pair ρ and $1 - \rho$, since

$$-\left(1 - \rho - \frac{1}{2}\right)^2 = -\left(\rho - \frac{1}{2}\right)^2.$$

Complex conjugation acts by

$$m_\rho^2 \mapsto \overline{m_\rho^2}.$$

We equip Σ_ζ with its natural locally finite divisor topology.

Definition 49 (Arithmetic mass divisor) Let $\xi(s)$ denote the completed Riemann zeta function, and let

$$Z_\xi := \{\rho \in \mathbb{C} : \xi(\rho) = 0, 0 < \Re(\rho) < 1\}$$

be the nontrivial zero divisor, counted with multiplicity. Define the shifted arithmetic mass-squared divisor

$$\Sigma_\zeta := \left\{ m_\rho^2 := -\left(\rho - \frac{1}{2}\right)^2 : \rho \in Z_\xi \right\}.$$

The functional equation identifies the pair ρ and $1 - \rho$, since

$$-\left(1 - \rho - \frac{1}{2}\right)^2 = -\left(\rho - \frac{1}{2}\right)^2.$$

Complex conjugation acts by

$$m_\rho^2 \mapsto \overline{m_\rho^2}.$$

We equip Σ_ζ with its natural locally finite divisor topology.

Definition 50 (UCFT spectral algebra) Let $\phi_0 \in \text{EIII}$ be a vacuum and let

$$\Delta_{\text{UCFT}} = \nabla^* \nabla + \mathcal{R} + \text{Hess}_{\phi_0}(V)$$

be the quadratic fluctuation operator in the universal coset field theory. After the required gauge fixing and removal of zero modes, take Δ_{UCFT} to be a self-adjoint nonnegative operator on a Hilbert space $\mathcal{H}_{\text{UCFT}}$. Let

$$\Sigma_{\text{UCFT}} := \text{Spec}(\Delta_{\text{UCFT}}) \subseteq \mathbb{R}_{\geq 0}.$$

Define the commutative spectral C^* -algebra

$$\mathcal{A}_{\text{UCFT}} := C_0(\Sigma_{\text{UCFT}}).$$

Similarly define

$$\mathcal{A}_{\zeta} := C_0(\Sigma_{\zeta}),$$

where Σ_{ζ} is the arithmetic mass-squared divisor.

Definition 51 (Galois-equivariant spectral realization) A Galois-equivariant spectral realization of the zeta sector in the UCFT sector is a $*$ -isomorphism of commutative spectral C^* -algebras

$$\Theta : \mathcal{A}_{\zeta} \xrightarrow{\sim} \mathcal{A}_{\text{UCFT}}$$

such that:

1. Θ preserves the distinguished mass-squared coordinate;
2. Θ intertwines complex conjugation on Σ_{ζ} with the real structure of the UCFT fluctuation Hilbert space;
3. at every finite algebraic approximation level, Θ is compatible with the relevant Galois action on the arithmetic mass divisor and the induced algebraic symmetry action on the UCFT spectral data.

Theorem 52 (Albert–EIII spectral criterion for the Riemann hypothesis) *For every Galois-equivariant spectral realization*

$$\Theta : \mathcal{A}_{\zeta} \xrightarrow{\sim} \mathcal{A}_{\text{UCFT}},$$

the Riemann hypothesis holds.

Proof Since

$$\Theta : \mathcal{A}_{\zeta} \xrightarrow{\sim} \mathcal{A}_{\text{UCFT}}$$

is a $*$ -isomorphism of commutative C^* -algebras, Gelfand duality gives a homeomorphism of spectra

$$\Theta^{\vee} : \Sigma_{\text{UCFT}} \xrightarrow{\sim} \Sigma_{\zeta}.$$

The UCFT fluctuation operator is self-adjoint and nonnegative, hence

$$\Sigma_{\text{UCFT}} \subseteq \mathbb{R}_{\geq 0}.$$

Because the realization preserves the distinguished mass-squared coordinate, the transported arithmetic mass-squared divisor also satisfies

$$\Sigma_{\zeta} \subseteq \mathbb{R}_{\geq 0}.$$

Let $\rho \in Z_{\zeta}$. Write

$$\rho = \frac{1}{2} + \alpha + i\gamma, \quad \alpha, \gamma \in \mathbb{R}.$$

Then

$$m_\rho^2 = -\left(\rho - \frac{1}{2}\right)^2 = -(\alpha + i\gamma)^2 = \gamma^2 - \alpha^2 - 2i\alpha\gamma.$$

Since $m_\rho^2 \in \mathbb{R}_{\geq 0}$, its imaginary part vanishes:

$$-2\alpha\gamma = 0.$$

For a nontrivial zero of the completed Riemann zeta function in the critical strip, $\gamma \neq 0$. Hence

$$\alpha = 0.$$

Thus

$$\Re(\rho) = \frac{1}{2}.$$

Since this holds for every nontrivial zero ρ , the Riemann hypothesis follows. $\square$

Definition 53 (Locally finite divisor topology) Let $\text{Div}_{\text{loc}}(\mathbb{C})$ denote the space of locally finite effective divisors on $\mathbb{C}$. Thus an element is a formal sum

$$D = \sum_{\lambda \in \mathbb{C}} m_\lambda [\lambda], \quad m_\lambda \in \mathbb{Z}_{\geq 0},$$

such that only finitely many λ 's occur in each compact subset of $\mathbb{C}$.

We equip $\text{Div}_{\text{loc}}(\mathbb{C})$ with the vague topology: a sequence D_n converges to D if and only if, for every compactly supported continuous test function $\varphi \in C_c(\mathbb{C})$,

$$\sum_{\lambda} m_{\lambda}^{(n)} \varphi(\lambda) \longrightarrow \sum_{\lambda} m_{\lambda} \varphi(\lambda).$$

Definition 54 (Spectral continuity) Let B be a parameter space containing a coset point b_{UCFT} and an arithmetic point b_{ζ} . A spectral bridge from the UCFT coset sector to the zeta sector is a family of closed operators

$$\{\mathcal{M}_b^2\}_{b \in B}$$

such that:

1. each $\mathcal{M}_b^2$ acts on a Hilbert space $\mathcal{H}_b$;
2. the family is continuous in the norm-resolvent topology, meaning that for some, equivalently every, $w \in \mathbb{C} \setminus \mathbb{R}$,

$$b \longmapsto (\mathcal{M}_b^2 - w)^{-1}$$

is norm-continuous;

3. each $\mathcal{M}_b^2$ admits a regularized determinant in the mass-squared coordinate,

$$D_b(\mu) := \det_{\text{reg}}(\mathcal{M}_b^2 - \mu);$$

4. the zero-divisor map

$$b \longmapsto \text{div}(D_b)$$

is continuous as a map

$$B \longrightarrow \text{Div}_{\text{loc}}(\mathbb{C}).$$

Definition 55 (Galois-compatible spectral bridge) A spectral bridge is Galois-compatible if, at every finite algebraic approximation level, the divisor

$$\operatorname{div}(D_b)$$

is stable under the relevant Galois action, and the transition maps in the bridge intertwine this Galois action with the real structure on the quantum Hilbert spaces $\mathcal{H}_b$.

Equivalently, for every admissible finite approximation $D_{b,N}$, every Galois element σ , and every point λ of the divisor, one has

$$\sigma(\lambda) \in \operatorname{supp}(D_{b,N})$$

with the same multiplicity, and this compatibility is preserved under passage to the limiting divisor.

Definition 56 (Positive continuous Albert–zeta realization) A positive continuous Albert–zeta realization is a Galois-compatible spectral bridge

$$\{\mathcal{M}_b^2\}_{b \in B}$$

such that:

1. the UCFT endpoint satisfies

$$\mathcal{M}_{b_{\text{UCFT}}}^2 = \Delta_{\text{UCFT}};$$

2. the arithmetic endpoint satisfies

$$\operatorname{div}(D_{b_\zeta}) = \Sigma_\zeta = \left\{ -\left(\rho - \frac{1}{2}\right)^2 : \xi(\rho) = 0, 0 < \Re(\rho) < 1 \right\};$$

3. every operator in the bridge is self-adjoint and nonnegative:

$$\mathcal{M}_b^2 = (\mathcal{M}_b^2)^*, \quad \mathcal{M}_b^2 \geq 0$$

for all $b \in B$.

Theorem 57 (Continuous spectral criterion for the Riemann hypothesis) *Every positive continuous Albert–zeta realization implies the Riemann hypothesis.*

Proof For each $b \in B$, self-adjointness implies

$$\operatorname{Spec}(\mathcal{M}_b^2) \subseteq \mathbb{R}.$$

Nonnegativity strengthens this to

$$\operatorname{Spec}(\mathcal{M}_b^2) \subseteq \mathbb{R}_{\geq 0}.$$

Since

$$D_b(\mu) = \det_{\operatorname{reg}}(\mathcal{M}_b^2 - \mu),$$

the zero divisor of D_b lies in the mass-squared spectrum:

$$\operatorname{div}(D_b) \subseteq \mathbb{R}_{\geq 0}.$$

At the arithmetic endpoint $b = b_\zeta$, the zero divisor is

$$\Sigma_\zeta = \left\{ -\left(\rho - \frac{1}{2}\right)^2 : \xi(\rho) = 0, 0 < \Re(\rho) < 1 \right\}.$$

Thus every shifted arithmetic mass-squared value is real and nonnegative.

Let

$$\rho = \frac{1}{2} + \alpha + i\gamma, \quad \alpha, \gamma \in \mathbb{R}.$$

Then

$$-\left(\rho - \frac{1}{2}\right)^2 = -(\alpha + i\gamma)^2 = \gamma^2 - \alpha^2 - 2i\alpha\gamma.$$

Since this number is real, its imaginary part vanishes:

$$-2\alpha\gamma = 0.$$

A nontrivial zero of $\xi(s)$ in the critical strip is not real, so

$$\gamma \neq 0.$$

Hence

$$\alpha = 0.$$

Therefore

$$\Re(\rho) = \frac{1}{2}.$$

Since this holds for every nontrivial zero, the Riemann hypothesis follows. $\square$

7 Heterotic Interface and Local Matching

The construction developed above realizes gravity and spacetime as *induced* structures: there is no fundamental graviton, and the metric $g_{\mu\nu}$ is the pull-back of the EIII metric along a spacetime-filling configuration ([Theorem 18](#)), while the Planck mass is generated radiatively in the sense of Sakharov and Adler [[6](#), [7](#)]. This motivates a precise, but limited, comparison between the coset sector and the massless bosonic sector of an $E_8 \times E_8$ heterotic compactification [[24](#)] whose visible gauge group is E_6 . We emphasize at the outset that the statement established here is local, two-derivative, and current-algebraic; it is an interface with a heterotic visible sector, not a proof of an exact holographic duality.

7.1 The heterotic datum

Fix an $E_8 \times E_8$ heterotic background on $\mathbb{R}^{1,3} \times X$ with X a six-dimensional internal space whose worldsheet realization preserves an E_6 current algebra at level one, so that the visible massless spectrum organizes into E_6 multiplets and the 78 adjoint is present as a gauge multiplet. Let M_{res} denote the residual scalar target obtained after the internal CFT has fixed all moduli except those of the visible E_6 -breaking sector. We write $H = \text{SO}(10) \times \text{U}(1)_\psi$ for the grand-unified isotropy group singled out by an adjoint Higgs vacuum in the $E_6 \supset H$ singlet direction ([section 8](#)).

Definition 58 (Interface map) The *interface map* is the E_6 -equivariant map

$$F : M_{\text{res}} \longrightarrow \mathcal{M} = E_6/(\text{SO}(10) \times \text{U}(1))$$

sending the gauge-unfixed adjoint-Higgs orbit of the heterotic model onto $\mathcal{M}$, and identifying the residual scalar fluctuations with the Goldstone coordinates ϕ^α of [section 3](#).

Lemma 59 (Two-derivative matching) *On the image of F the heterotic non-linear sigma-model metric restricted to the visible E_6 -breaking directions coincides, up to an overall normalization f^2 , with the pull-back $F^*\mathfrak{g}$ of the unique E_6 -invariant EIII metric $\mathfrak{g}_{\alpha\beta} = f^2(-K_{\alpha\beta})$.*

Proof The residual visible directions transform in $16_{-3} \oplus \overline{16}_{+3}$ under H , which is real-irreducible (Theorem 18). Any H -invariant symmetric two-tensor on this representation is unique up to scale by Schur's lemma; both the sigma-model metric and $\mathfrak{g}$ are such tensors, hence proportional. The proportionality constant defines f^2 . $\square$

Lemma 60 (Current-algebra matching) *The $H = \mathrm{SO}(10) \times \mathrm{U}(1)_\psi$ isometry action on $\mathcal{M}$ is generated by the same currents as the level-one $\mathrm{SO}(10)$ and $\mathrm{U}(1)_\psi$ Kač–Moody currents of the heterotic worldsheet theory, with matched central data. In particular the $\mathrm{U}(1)_\psi$ charge assignment $16_{-3} \oplus \overline{16}_{+3}$ reproduces the worldsheet charge lattice of the visible matter.*

Proof The Killing vector fields of $\mathcal{M}$ generating H act on the tangent representation exactly as the zero modes of the $\mathrm{SO}(10) \times \mathrm{U}(1)_\psi$ currents act on the visible states; the charges agree by construction of F , and the $\mathrm{U}(1)_\psi$ normalization is fixed by requiring the 16 to carry charge -3 , matching the branching $78 = 45_0 \oplus 1_0 \oplus 16_{-3} \oplus \overline{16}_{+3}$. $\square$

Theorem 61 (Local heterotic realization) *Let F be the interface map of Definition 58. Then, in a neighbourhood of the gauge-unfixed Higgs orbit, the two-derivative bosonic action of the visible E_6 -breaking sector of the heterotic model is mapped by F onto the coset action of section 3, with the induced Einstein–Hilbert term of section 4 providing the gravitational sector. Under this map the Green–Schwarz axion [25] of the heterotic background is identified with the $\mathrm{U}(1)_\psi$ Goldstone mode of $\mathcal{M}$.*

Proof Combine Lemma 59 and Lemma 60: the kinetic term matches by two-derivative matching and the symmetry action matches by current-algebra matching, so the visible bosonic Lagrangians agree to two derivatives. The gravitational sector is not present in the heterotic two-derivative visible action as an independent field; on the side it is induced (section 4), consistent with the absence of a fundamental visible-sector graviton. The Green–Schwarz axion couples to $\mathrm{Tr} F \wedge F$ with a shift symmetry, and the interface identifies the corresponding visible shift mode with the $\mathrm{U}(1)_\psi$ Goldstone of $\mathcal{M}$. $\square$

Remark 62 (Scope of the correspondence) Theorem 61 is a statement about the two-derivative bosonic action and the current algebra only. It does not assert an exact conformal duality, a matching of the full string spectrum, or agreement of higher-derivative α' corrections. In particular the heterotic side carries an infinite tower of massive string states with no coset counterpart, and the coset side is an effective field theory valid below the string scale. The value of the correspondence is structural: it exhibits the coset as the low-energy image of a genuine string background, and it shows that the induced-gravity mechanism is compatible with a formulation in which no fundamental visible-sector graviton is postulated.

Table 2 Interface dictionary between the visible heterotic sector and the coset sector. The correspondence is established at the level of the two-derivative bosonic action and the $\text{SO}(10) \times \text{U}(1)_\psi$ current algebra.

Heterotic $E_8 \times E_8$ visible sector	coset sector
E_6 current algebra (level one)	Compact E_6 symmetry selected by $J_3(\mathbb{O})$
Adjoint Higgs orbit $E_6 \cdot \langle 78_H \rangle$	Vacuum manifold $\mathcal{M} = E_6/(\text{SO}(10) \times \text{U}(1))$
Visible matter $16_{-3} \oplus \overline{16}_{+3}$	Goldstone tangent space $T\mathcal{M}$
Sigma-model metric on visible moduli	EIII metric $\mathfrak{g} = f^2(-K)$
$\text{SO}(10) \times \text{U}(1)_\psi$ Kač–Moody currents	Isometry generators of $\mathcal{M}$
Green–Schwarz axion	$\text{U}(1)_\psi$ Goldstone mode
No fundamental visible graviton	Induced (Sakharov–Adler) gravity

8 The $\mathbb{Z}_{12}$ Asymmetric-Orbifold Realization

We now consider a heterotic compactification intended to meet the criteria of [section 7](#): a $\mathbb{Z}_{12}$ asymmetric orbifold of the $E_8 \times E_8$ heterotic string [\[26, 27\]](#) whose visible sector is a three-generation E_6 grand unified theory with an adjoint Higgs 78_H [\[28\]](#). Giving the adjoint Higgs a vacuum expectation value in the $E_6 \supset H$ singlet direction breaks $E_6 \rightarrow H = \text{SO}(10) \times \text{U}(1)_\psi$, and the local orbit computation identifies the resulting gauge-unfixed Higgs orbit with the coset $\mathcal{M}$.

8.1 The orbifold datum

The model is a $\mathbb{Z}_{12}$ asymmetric orbifold with visible gauge group E_6 realized at level one, three chiral generations in the 27 , and an adjoint Higgs multiplet 78_H arising from the diagonal embedding of the visible E_6 in the ten-dimensional gauge group. Under $E_6 \supset H = \text{SO}(10) \times \text{U}(1)_\psi$ the adjoint branches as

$$78 = 45_0 \oplus 1_0 \oplus 16_{-3} \oplus \overline{16}_{+3}, \quad (13)$$

and we take the Higgs vacuum $\langle 78_H \rangle = \Phi_0$ to lie in the 1_0 direction.

Theorem 63 (Gauge-unfixed EIII orbit theorem) *Let Φ_0 be an adjoint-Higgs vacuum in the $1_0 \subset 78$ direction under $E_6 \supset H = \text{SO}(10) \times \text{U}(1)_\psi$. Then the gauge-unfixed Higgs orbit is*

$$\mathcal{O}_H = E_6 \cdot \Phi_0 \cong E_6/(\text{SO}(10) \times \text{U}(1)) = \mathcal{M}, \quad T_{\mathbb{C}} \mathcal{O}_H = 16_{-3} \oplus \overline{16}_{+3}.$$

Proof By the branching [\(13\)](#) the centralizer in E_6 of a generic vector in the 1_0 direction is exactly $H = \text{SO}(10) \times \text{U}(1)_\psi$: the 45_0 commutes with Φ_0 , the $\text{U}(1)_\psi$ generator is Φ_0 itself, and the $16_{-3} \oplus \overline{16}_{+3}$ carry non-zero $\text{U}(1)_\psi$ charge and are therefore rotated. By the orbit-stabilizer theorem $E_6 \cdot \Phi_0 \cong E_6/H = E_6/(\text{SO}(10) \times \text{U}(1))$. The tangent space at Φ_0 is the span of the broken generators, namely $16_{-3} \oplus \overline{16}_{+3}$, giving the stated complexified tangent representation. This is the coset $\mathcal{M}$ of [section 3](#). $\square$

Theorem 64 (Gauge-fixed/gauge-unfixed equivalence) *The EIII scalar sector is physically equivalent to the gauge-unfixed Stueckelberg/Higgs-orbit description of the $E_6 \rightarrow \text{SO}(10) \times \text{U}(1)_\psi$ breaking sector of the $\mathbb{Z}_{12}$ model. Unitary gauge maps this description to the standard massive-vector description of the heterotic E_6 grand unified theory, in which the $16_{-3} \oplus \overline{16}_{+3}$ Goldstones are absorbed by the broken gauge bosons.*

Proof In the gauge-unfixed description the 32 real coordinates of $\mathcal{O}_H \cong \mathcal{M}$ are Stueckelberg/Goldstone fields for the broken E_6/H directions. Fixing unitary gauge sets the Goldstones to zero and gives mass to the corresponding gauge bosons via the Higgs mechanism; the massive vector content is gauge-invariant and hence the same in both descriptions. The coset action of [section 3](#) is the gauge-unfixed Goldstone dynamics on $\mathcal{O}_H$, so the two descriptions are equivalent at the level of this sector. $\square$

Corollary 65 (Breaking chain) *The $\mathbb{Z}_{12}$ model realizes the phenomenological breaking chain*

$$E_6 \rightarrow \text{SO}(10) \times \text{U}(1)_\psi \rightarrow \text{SU}(5) \rightarrow \text{SU}(3) \times \text{SU}(2) \times \text{U}(1),$$

with the first step realized geometrically as the emergence of the coset $\mathcal{M}$ and the subsequent steps by the standard heterotic Higgs mechanism.

Proof The first arrow is [Theorem 63](#). The remaining arrows are the standard $\text{SO}(10) \rightarrow \text{SU}(5) \rightarrow \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ descent of a heterotic E_6 grand unified theory with the matter content of the 27; three generations follow from the three fixed-point families of the $\mathbb{Z}_{12}$ action [\[28\]](#). $\square$

Corollary 66 (UV completion) *If the $\mathbb{Z}_{12}$ model of [\[28\]](#) satisfies the global heterotic consistency conditions listed in [Remark 68](#), then it is an ultraviolet completion of the effective theory in the sense of the local correspondence of [section 7](#).*

Proof Immediate from [Theorem 63](#), [Theorem 61](#), and the definition of the global heterotic consistency conditions: local matching plus global worldsheet consistency is the required condition for the heterotic model to define a consistent string background whose low-energy visible sector is the coset theory. $\square$

9 The Albert–EIII Swampland Criterion

The results above give a *relative* swampland criterion. We do not claim that all non-coset vacua are inconsistent. The statement is restricted to the Albert-selected heterotic class: within that class, a vacuum can realize the coset correspondence only when its gauge-unfixed Higgs/moduli sector is EIII with the data specified below.

Theorem 67 (Albert–EIII swampland theorem) *Let $\mathcal{T}$ be a four-dimensional effective theory proposed as the Albert-selected low-energy sector of an $E_8 \times E_8$ heterotic compactification. Assume that its visible exceptional sector is selected by the Albert algebra $J_3(\mathbb{O})$ and that the*

corresponding compact real form is E_6 . Then $\mathcal{T}$ lies in the Albert-selected heterotic landscape only if its gauge-unfixed Higgs/moduli sector contains a residual scalar orbit

$$\mathcal{O} \cong E_6/(\mathrm{SO}(10) \times \mathrm{U}(1)) = \mathcal{M}, \quad T_{\mathbb{C}} \mathcal{O} = 16_{-3} \oplus \overline{16}_{+3},$$

carrying the unique positive E_6 -invariant EIII metric, globally consistent $\mathrm{SO}(10) \times \mathrm{U}(1)_{\psi}$ gauge data, anomaly cancellation, spin-lift compatibility, Green–Schwarz matching, and heterotic worldsheet consistency. Equivalently, any proposed Albert-selected four-dimensional effective theory failing one of these conditions lies outside the Albert–EIII landscape: it has no $E_8 \times E_8$ heterotic ultraviolet completion of the specified type realizing the coset correspondence.

Proof The argument is by necessity together with the contrapositive.

Type and real form. Albert selection fixes the complex exceptional type to E_6 (section 2); choosing the compact real form gives the compact E_6 structure used in the EIII target and guarantees a positive-definite invariant metric (Theorem 24).

Forced orbit. Imposing the grand-unified isotropy $H = \mathrm{SO}(10) \times \mathrm{U}(1)_{\psi}$ forces the homogeneous scalar orbit to be E_6/H . By the orbit–stabilizer theorem an adjoint Higgs vacuum in the 1_0 direction of the branching (13) has centralizer H , so its gauge-unfixed Higgs orbit is exactly $E_6 \cdot \Phi_0 \cong E_6/(\mathrm{SO}(10) \times \mathrm{U}(1))$ (Theorem 63).

Rigidity. The tangent representation is forced to be $16_{-3} \oplus \overline{16}_{+3}$, which is real-irreducible. Hence the H -invariant positive metric is unique up to scale, and there is no competing compact homogeneous residual scalar sector with the same Albert/ E_6/H data (Theorem 18).

No-soldering exclusion. Because $\mathfrak{m}$ has no invariant four-plane and the EIII metric is positive definite, the coset cannot supply a Lorentzian vierbein by selecting four invariant scalar directions. The valid construction is the nondegenerate Euclidean pull-back of Theorem 18, followed by Osterwalder–Schrader reconstruction; gravity then enters through the induced Sakharov–Adler dynamics of section 4.

Global constraints. Local EIII matching is not sufficient; a heterotic completion must further satisfy modular invariance, GSO/fermion-parity consistency, flux and Wilson-line quantization, bundle transition-function compatibility, anomaly cancellation on overlaps, spin-lift conditions for the 16 matter, and Green–Schwarz matching [25].

Contrapositive. If a proposed effective theory lacks the EIII orbit, the required tangent representation, the EIII metric, the correct gauge action, the anomaly/Green–Schwarz compatibility, the spin-lift data, or global heterotic consistency, then it cannot be the Albert-selected $E_8 \times E_8$ ultraviolet completion of the specified type. $\square$

Remark 68 (Global heterotic consistency ledger) The global heterotic consistency conditions invoked above and below are: modular invariance, GSO/fermion-parity consistency, flux and Wilson-line quantization, bundle transition-function compatibility, anomaly cancellation on overlaps, spin-lift conditions for the 16 matter, and Green–Schwarz matching [25].

Corollary 69 ($\mathbb{Z}_{12}$ test) *A $\mathbb{Z}_{12}$ asymmetric heterotic E_6 orbifold realizes the coset correspondence if its gauge-unfixed adjoint-Higgs orbit is $E_6/(\mathrm{SO}(10) \times \mathrm{U}(1))$ and if its full worldsheet, anomaly, bundle, flux, spectrum, renormalization-group, and gravitational matching data satisfy the global heterotic consistency conditions of Remark 68. If these checks fail, the model remains an E_6 -aligned heterotic model but not the coset ultraviolet completion.*

Proof The orbit condition is [Theorem 63](#) and places the model in the Albert-selected landscape of [Theorem 67](#); the global conditions are those of [Remark 68](#) and [Corollary 66](#). If any global condition fails, the contrapositive of [Theorem 67](#) applies. $\square$

Corollary 70 ($\mathbb{Z}_{12}$ candidate corollary) *The $\mathbb{Z}_{12}$ asymmetric E_6 orbifold is a concrete candidate for realizing the coset correspondence because it supplies an $E_8 \times E_8$ heterotic origin, a visible E_6 grand unified sector, three generations, and an adjoint Higgs obtained by diagonal embedding. If its gauge-unfixed adjoint-Higgs orbit, anomaly data, worldsheet data, and gravitational reduction satisfy the global heterotic equivalence criterion of [Remark 68](#), then it gives a heterotic ultraviolet completion of the coset theory.*

Proof The four structural features are those of the model of [\[28\]](#) recorded in [section 8](#); each is a prerequisite of [Theorem 67](#), and no other presently identified construction is known to supply all four together with the EIII orbit of [Theorem 63](#). $\square$

Remark 71 (Status of the criterion) [Theorem 67](#) is a relative, class-restricted statement: it constrains only theories proposed as Albert-selected heterotic vacua. It is not a universal assertion that all other four-dimensional effective theories are inconsistent. Within its stated scope it is sharp, and it makes the $\mathbb{Z}_{12}$ model of [section 8](#) a concrete landscape candidate subject to the checklist of [Corollary 69](#).

Declarations

The author declares no competing interests. No datasets were generated or analyzed in this work. No external funding is reported.

Appendix A Albert Algebra Proofs

This appendix supplies proofs regarding the Albert algebra deferred from [section 2](#).

A.1 Dimension and Jordan Identity

Proof of [Theorem 6](#) Choose a basis $\{e_0, e_1, \dots, e_7\}$ of the octonions satisfying $e_0 = 1$ and $e_i^2 = -1$ for $i \geq 1$. Every $X \in J_3(\mathbb{O})$ can be written uniquely as the Hermitian 3×3 matrix

$$X = \begin{pmatrix} \alpha_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \alpha_2 & x_1 \\ x_2 & \bar{x}_1 & \alpha_3 \end{pmatrix}, \quad \alpha_i \in \mathbb{R}, \quad x_i \in \mathbb{O}.$$

The real-dimension count is therefore 3 scalars $+ 3 \times 8$ octonion components $= 27$. Jordan commutativity is obvious from the symmetric product $\circ$. The Jordan identity $a \circ (b \circ a^2) = (a \circ b) \circ a^2$ holds for every special Jordan algebra; it follows from the alternative law of the octonions [\[3\]](#). Hence $J_3(\mathbb{O})$ is indeed a 27-dimensional Jordan algebra. $\square$

A.2 Quadratic and Cubic Invariants

Proof of Theorem 7 For X as above set

$$Q(X) = \text{Tr}(X^2), \quad N(X) = \alpha_1 \alpha_2 \alpha_3 - \sum_{i=1}^3 \alpha_i \|x_i\|^2 + 2 \Re(x_1 x_2 x_3),$$

where juxtaposition is octonion multiplication. The form Q is quadratic and N is cubic in the real coordinates. Direct calculation shows $Q(g \cdot X) = Q(X)$ and $N(g \cdot X) = \lambda(g)N(X)$ for every $g \in \text{Str}(J_3(\mathbb{O}))$ with multiplicative character $\lambda(g) \in \mathbb{R}^\times$ [1]. In particular, the subgroup with $\lambda(g) = 1$ preserves both forms. $\square$

A.3 Exceptional Type E_6

Proof of Theorem 8 Let $\text{Str}^0(J_3(\mathbb{O}))$ denote the connected component of the identity in $\text{Str}(J_3(\mathbb{O}))$. Jacobson's construction [1] realizes the exceptional structure algebra as the sum of derivations of $J_3(\mathbb{O})$ (an $\mathfrak{f}_4$) plus the traceless right multiplications

$$R_X : Y \mapsto X \circ Y - \frac{1}{3} Q(X, Y) \mathbf{1}.$$

After complexification this Lie algebra has type $\mathfrak{e}_6$. The main text uses the simply connected compact real form of the same complex Lie type for the EIII homogeneous space; this is the group denoted E_6 throughout the coset construction. $\square$

A.4 Orbit Classification

Lemma 72 *If $g \in E_6$ and $X \in J_3(\mathbb{O})$ then $Q(g \cdot X) = Q(X)$ and $N(g \cdot X) = N(X)$.*

Proof The compact E_6 action used on the regular stratum preserves the restricted quadratic and cubic invariants, so the claim follows from Theorem 7. $\square$

Proof of Theorem 9 Suppose $g \cdot X = Y$. Lemma 72 gives equality of (Q, N) . Conversely, restrict to the regular compact stratum specified in the theorem. The standard invariant-theoretic description of the compact E_6 action [2] shows that the common level set of (Q, N) through X_0 is a single compact orbit. Hence two points on that level set differ by an element of E_6 . $\square$

A.5 An $\text{SO}(10) \times \text{U}(1)$ Stabilizer

Proof of Corollary 10 The compact group of type E_6 has the maximal-rank subgroup $\text{Spin}(10) \times \text{U}(1)$ [4]. Choosing X_0 in the corresponding regular orbit gives stabilizer Lie algebra $\mathfrak{so}(10) \oplus \mathfrak{u}(1)$ of dimension $45 + 1 = 46$. With the global convention used in this paper, the stabilizer is denoted $\text{SO}(10) \times \text{U}(1)$. $\square$

A.6 Coercivity of the Symmetry-Breaking Potential

Lemma 73 *For $\alpha > 0$, $k \geq 2$ even, the polynomial $V(X) = P(X) + \alpha[I_2(X) - c_2]^k$ satisfies $V(X) \rightarrow \infty$ as $\|X\| \rightarrow \infty$.*

Proof Write $I_2(X) = Q(X) = \beta \|X\|^2 + O(\|X\|)$ for some $\beta > 0$. Since k is even,

$$\alpha[I_2(X) - c_2]^k = \alpha\beta^k \|X\|^{2k} + O(\|X\|^{2k-1}),$$

so V is radially unbounded. $\square$

Lemma 74 $P(X) \geq 0$ for all $X \in J_3(\mathbb{O})$, with equality if and only if $X \in E_6 \cdot X_0$ on the regular stratum.

Proof Non-negativity is clear from (1). Equality implies

$$I_2(X) = I_2(X_0) \quad \text{and} \quad I_3(X) = I_3(X_0);$$

[Theorem 9](#) then yields the stated orbit condition. $\square$

Appendix B Emergent Spacetime Proofs

This appendix supplies proofs for [Lemma 17](#) and [Theorem 18](#) from [section 3](#).

B.1 Rank-Four Embedding

Recall that the broken directions of the coset

$$\mathcal{M} = \frac{E_6}{\mathrm{SO}(10) \times \mathrm{U}(1)}$$

form a 32-dimensional real vector space $\mathfrak{m}$ carrying the unique invariant metric $\mathfrak{g}_{\alpha\beta} = f^2(-K_{\alpha\beta})$ of [Theorem 24](#), which on the compact real form of E_6 is positive definite. Let

$$d\phi_x : T_x \mathbb{R}^4 \rightarrow T_{\phi(x)} \mathcal{M} \cong \mathfrak{m}$$

denote the differential of the map $\phi^\alpha(x)$.

Proof of Lemma 17 The pull-back at x is the quadratic form

$$g_{\mu\nu}(x) = \mathfrak{g}_{\alpha\beta} \partial_\mu \phi^\alpha \partial_\nu \phi^\beta = \mathfrak{g}(d\phi_x(\partial_\mu), d\phi_x(\partial_\nu)).$$

Because $\mathfrak{g}$ is positive definite, the associated Gram map obeys $\mathrm{rank} \, g_{\mu\nu} = \mathrm{rank} \, d\phi_x = \dim V_x$, where $V_x = \Im d\phi_x \subseteq \mathfrak{m}$: indeed $g_{\mu\nu} v^\mu = 0$ iff $\mathfrak{g}(d\phi_x v, \cdot) = 0$ iff $d\phi_x v = 0$ by definiteness. Since $\dim T_x \mathbb{R}^4 = 4$ we have $\dim V_x \leq 4$, and non-degeneracy of $g_{\mu\nu}$ forces $\dim V_x = 4$, i.e. $d\phi_x$ is injective and $V_x = \mathfrak{p}_0(x)$ is a four-plane. Its $\mathfrak{g}$ -orthogonal complement $\mathfrak{p}_m(x)$ has dimension 28. $\square$

B.2 Euclidean Signature and Continuation

Proof of Theorem 18 Embed $\mathrm{SO}(10) \times \mathrm{U}(1)$ in E_6 via the stabilizer of $X_0 = \mathrm{diag}(1, 1, 1) \in J_3(\mathbb{O})$, and write $\mathfrak{e}_6 = \mathfrak{h} \oplus \mathfrak{m}$ with $\mathfrak{h} = \mathfrak{so}(10) \oplus \mathfrak{u}(1)$. On the *compact* real form of E_6 the Killing form is negative definite on all of $\mathfrak{e}_6$; hence $-K$, and therefore $\mathfrak{g} = f^2(-K)$, is positive definite on $\mathfrak{m}$. (This is the same fact that makes the symmetric space $\mathcal{M} = \frac{E_6}{\mathrm{SO}(10) \times \mathrm{U}(1)}$ Riemannian; EIII is Hermitian symmetric, hence Riemannian, and carries no invariant indefinite metric.)

By Lemma 17 the tetrad $e_\mu^a = \theta_\alpha^a \partial_\mu \phi^\alpha$ is invertible for a non-degenerate configuration. Choosing a $\mathfrak{g}$ -orthonormal basis $\{u_a\}_{a=0}^3$ of $\mathfrak{p}_0(x)$, so that $\mathfrak{g}(u_a, u_b) = \delta_{ab}$, and decomposing $\partial_\mu \phi^\alpha = e_\mu^a u_a^\alpha$,

$$g_{\mu\nu} = \mathfrak{g}(u_a, u_b) e_\mu^a e_\nu^b = \delta_{ab} e_\mu^a e_\nu^b,$$

which is symmetric, non-degenerate (invertibility of e_μ^a), and positive definite. Smoothness in x follows from smoothness of ϕ and of the Maurer-Cartan form. Openness and density of the non-degenerate locus hold because $\det e_\mu^a \neq 0$ is an open condition satisfied by generic immersions. Thus $g_{\mu\nu}$ is a Riemannian metric on $\mathbb{R}^4$, namely the first fundamental form of ϕ .

The signature is $(4, 0)$, not $(1, 3)$: a pull-back of a positive-definite form is positive semidefinite for every ϕ , so no choice of configuration can produce a Lorentzian induced metric. $\square$

Appendix C Heat-Kernel Coefficients and Induced Gravity

C.1 Invariant Metric on the Coset

Proof of Theorem 24 Write the Cartan decomposition $\mathfrak{e}_6 = \mathfrak{h} \oplus \mathfrak{m}$ with $\mathfrak{h} = \mathfrak{so}(10) \oplus \mathfrak{u}(1)$. For $g \in E_6$ define the adjoint action $\text{Ad}_g(X) = gXg^{-1}$. Because $\mathfrak{m}$ transforms in an irreducible representation of $\text{SO}(10) \times \text{U}(1)$, Schur's lemma implies that any E_6 -invariant symmetric bilinear form on $\mathfrak{m}$ is proportional to the restriction of the Killing form $\mathfrak{g}_{\alpha\beta} = c K_{\alpha\beta}$. On the compact real form of E_6 the Killing form is negative definite, so $K|_{\mathfrak{m}} < 0$; positivity of the σ -model kinetic term therefore fixes $c = -f^2 < 0$, i.e. $\mathfrak{g}_{\alpha\beta} = f^2(-K_{\alpha\beta}) > 0$, completing the proof. $\square$

C.2 BRST Complex and Functional Determinants

Gauge fixing for the local $H = \text{SO}(10) \times \text{U}(1)$ symmetry follows the Faddeev-Popov procedure [13]. The collective matter multiplet is $\Phi = (\phi^\alpha, A_\mu^i, \psi)$, with $\alpha \in \{1, \dots, 32\}$ and $i \in \{1, \dots, 46\}$. Here, ϕ^α are the real coset-scalar fluctuations, A_μ^i the gauge-field fluctuations, and ψ the Weyl fermions. Ghost, antighost, and Nakanishi-Lautrup fields are denoted by $(c^i, \bar{c}^i, B^i)$. With the background Lorenz gauge $\mathcal{F}_i = \nabla^\mu A_{\mu i}$, the gauge-fixed action reads

$$\begin{aligned} S_{\text{BRST}} &= S[\Phi] + \int d^4x \sqrt{-g} \left[B^i \mathcal{F}_i + \frac{\xi}{2} B^i B_i + \bar{c}^i \Delta_{\text{ghost}} c_i \right], \\ \Delta_{\text{ghost}} &= -\nabla^2 \delta_{ij} + R_{ij}, \quad R_{ij} = f_{ij}{}^k F_{\mu\nu k} \gamma^\mu \gamma^\nu. \end{aligned} \tag{C1}$$

The nilpotent BRST operator s acts as

$$\begin{aligned} s \phi^\alpha &= -c^i (t_i)^\alpha_\beta \phi^\beta, & s A_\mu^i &= D_\mu c^i, \\ D_\mu c^i &= \partial_\mu c^i + f_{jk}^i A_\mu^j c^k, & s \psi &= i c^i T_i \psi, \\ s c^i &= -\frac{1}{2} f_{jk}^i c^j c^k, & s \bar{c}^i &= B^i, \\ s B^i &= 0, \end{aligned}$$

so $s^2 = 0$ and physical correlators are gauge-independent. Here, B^i is the Nakanishi–Lautrup auxiliary field, and $\gamma^\mu = e_a^\mu \gamma^a$ are the curved-space Dirac matrices obeying $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$. Eliminating B^i by its algebraic equation of motion reproduces the usual gauge-fixing term $(2\xi)^{-1}(\mathcal{F}_i)^2$ and leaves the ghost operator Δ_{ghost} defined above.

The quadratic expansion of S_{BRST} about a background $\bar{\Phi}$ is recorded in (C2); its functional determinants supply the one-loop effective action used in the next subsection to derive the induced Einstein-Hilbert term.

C.3 One-Loop Induced Einstein-Hilbert Term

Proof of Theorem 27 Let $\bar{\Phi}(x)$ be a fixed classical background and define the full fluctuation multiplet

$$\Phi(x) = (\phi^\alpha(x), A_\mu^i(x), \psi^a(x), c^i(x), \bar{c}^i(x)),$$

with $\alpha \in \{1, \dots, 32\}$ and $i \in \{1, \dots, 46\}$. The ghosts $c^i, \bar{c}^i$ transform in the adjoint of H . Writing $\Phi = \bar{\Phi} + \varphi$, the gauge-fixed BRST action (C1) expands as

$$S_{\text{BRST}}[\bar{\Phi} + \varphi] = S_{\text{BRST}}[\bar{\Phi}] + S^{(2)}[\varphi] + O(\varphi^3),$$

with quadratic part

$$S^{(2)}[\varphi] = \frac{1}{2} \int d^4x \sqrt{-g} \varphi^T \Delta \varphi, \quad \Delta = \begin{pmatrix} -\nabla^2 \mathbf{1}_{32} + M_\phi^2 & 0 & 0 \\ 0 & -\nabla^2 \mathbf{1}_{46} + M_A^2 & 0 \\ 0 & 0 & i\not{V} \end{pmatrix}, \quad (\text{C2})$$

where $(M_\phi^2)_{\alpha\beta} = [\text{Hess } V(X_0)]_{\alpha\beta}$ and M_A^2 is the gauge-field endomorphism, which vanishes in the unbroken H phase. The background-field derivation of (C2) follows Abbott’s method [29]; heat-kernel methods are reviewed in [30].

Because Gaussian path integrals satisfy [15, 29]

$$\int D\varphi e^{-\frac{1}{2}\varphi^T \Delta \varphi} = (\det \Delta_\phi)^{-\frac{1}{2}}, \quad \int D\bar{c} Dc e^{\bar{c} \Delta_{\text{ghost}} c} = \det \Delta_{\text{ghost}},$$

the one-loop effective action is

$$\Gamma^{(1)} = \frac{1}{2} \log \det(-\Delta_\phi) - \log \det(-\Delta_{\text{ghost}}) + \dots$$

The overall minus sign for the ghost determinant matches the background-field convention of [29]. Using the proper-time representation [31],

$$\log \det(-\Delta) = - \int_{\mu^{-2}}^{\Lambda^{-2}} \frac{d\tau}{\tau} \text{Tr}[e^{-\tau\Delta}] = - \sum_{n=0}^{\infty} a_{2n}(\Delta) \frac{\Lambda^{2-2n} - \mu^{2-2n}}{2-2n},$$

and the Seeley–DeWitt coefficients of Table C1, the a_2 term from the $N_s = 32$ Goldstone scalars yields

$$\Gamma^{(1)} \supset \frac{1}{2} M_{\text{Pl}}^2 \int d^4x \sqrt{-g} R, \quad M_{\text{Pl}}^2 = \frac{N_s f^2}{(4\pi)^2} \log \frac{\Lambda^2}{\mu^2}.$$

□

Table C1 Lowest Seeley-DeWitt coefficients (Euclidean signature) for minimal Laplace-type operators $\Delta = -\nabla^2 + \mathcal{E}$. All curvature tensors refer to $g_{\mu\nu}$.

Field	a_0	a_2	a_4
Real scalar	1	$\frac{1}{6}R$	$\frac{1}{180}(R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - R_{\mu\nu}R^{\mu\nu}) + \frac{1}{72}R^2$
Weyl fermion	2	$-\frac{1}{6}R$	$\frac{7}{720}R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - \frac{1}{180}R_{\mu\nu}R^{\mu\nu} + \frac{1}{144}R^2$
Gauge vector	2	$\frac{1}{6}R$	$\frac{1}{30}R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - \frac{1}{30}R_{\mu\nu}R^{\mu\nu}$
Vector ghost (Grassmann)	2	$\frac{1}{6}R$	$-\frac{11}{180}R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} + \frac{1}{30}R_{\mu\nu}R^{\mu\nu} - \frac{1}{60}\square R$

C.4 Mass Spectrum of the Coset Scalars

Proof of Theorem 28 Expanding the one-loop effective action (8) about $X = X_0 + \varphi$, $\varphi \in \mathfrak{p} = T_{X_0}\mathcal{M}$, and retaining only $\mathcal{O}(\varphi^2)$ terms yields (9):

$$\Gamma^{(1)}[X_0 + \varphi] = \Gamma^{(1)}[X_0] + \frac{1}{2} \int d^4x \varphi^I [\text{Hess}(X_0)]_{IJ} \varphi^J + \dots,$$

where $I, J \in \{1, \dots, 32\}$ label a basis of $\mathfrak{p}$.

The tree potential V of Theorem 12 is E_6 -invariant and vanishes on the whole orbit $\mathcal{M}$; hence $\text{Hess}(X_0) = 0$ identically at tree level, and all 32 coset scalars are exact Goldstone bosons. The gauge and Yukawa couplings of (6) respect only $H = \text{SO}(10) \times \text{U}(1)$, so the one-loop effective potential V_{eff} is H -invariant but not E_6 -invariant; its Hessian therefore commutes with the H -action. Since $\mathfrak{m} \cong \mathfrak{p}$ carries an *irreducible* representation of H (Lemma 23), Schur's lemma implies

$$\text{Hess}_{\text{eff}}(X_0) = m^2 \mathbf{1}_{32}, \quad m^2 = f^2 \mu^2 > 0,$$

a single eigenvalue fixed by the FRG flow; no H -invariant $4 + 28$ splitting of the mass matrix exists. On a non-degenerate spacetime-filling background the four directions $\mathfrak{p}_0 = \Im d\phi$ of Lemma 17 are singled out by the frame condensate $\langle \partial_\mu \phi^a \rangle = e_\mu^a$; their gradients constitute the tetrad, which is promoted to the gravitational field by the induced Einstein-Hilbert term (Theorem 27) and does not propagate as an independent scalar. The remaining 28 transverse fluctuations $\varphi^{\hat{\alpha}} \in \mathfrak{p}_m$ obey $(\square + m^2)\varphi^{\hat{\alpha}} = 0$ with $m^2 = f^2 \mu^2$. $\square$

Appendix D Functional Renormalization-Group Details

D.1 Wetterich Equation and Regulator Choice

We adopt the background-field functional RG [32, 33]. For the coarse-grained effective action Γ_k the exact flow equation reads

$$\partial_k \Gamma_k = \frac{1}{2} \text{Tr} \left[\left(\Gamma_k^{(2)} + R_k \right)^{-1} \partial_k R_k \right],$$

where $\Gamma_k^{(2)}$ is the Hessian with respect to the collective fields $\Phi = (\phi^\alpha, A_\mu^i, \psi)$ and R_k is an IR regulator that suppresses modes with $p^2 \lesssim k^2$. We work in background Lorenz gauge for the $\text{SO}(10) \times \text{U}(1)$ gauge sector, add the corresponding Faddeev-Popov determinant, and impose standard BRST sources. The background-field identities then guarantee that Γ_k remains gauge invariant at every k .

All threshold functions entering the flow are evaluated with

$$R_k(p^2) = Z_k(k^2 - p^2) \theta(k^2 - p^2),$$

where Z_k is the wave-function renormalization of the field in question. Litim's choice maximizes the gap in the inverse propagator and yields analytic expressions for the threshold integrals while preserving locality of Γ_k [34].

Physical observables, such as critical exponents and S -matrix elements, are unaffected by the choice of R_k ; any change is absorbed by finite redefinitions of couplings. Explicit regulator-variation checks for the present truncation reproduce the standard $\sim 1\%$ scheme uncertainty.

D.2 Dimensionless Couplings and Projection Rules

The truncation employed in [section 5](#) keeps only the lowest-dimensional operators that mix under the RG flow,

$$\Gamma_k = \int d^4x \left[\frac{1}{2f_k^2} \partial_\mu \phi^\alpha \partial^\mu \phi_\alpha + \frac{1}{4g_k^2} F_{\mu\nu}^i F^{i\mu\nu} + \frac{y_k}{2} \phi^\alpha \bar{\psi} \psi \right] + \Gamma_k^{\text{grav}}[\eta_{\mu\nu}],$$

where ϕ^α denotes the 32 coset scalars, $F_{\mu\nu}^i$ the $\text{SO}(10) \times \text{U}(1)$ field strength and ψ the chiral matter multiplet of [Table 1](#). Higher-derivative and multi-scalar operators enter only at subleading order in the derivative expansion and do not affect the universal data extracted here. We write $t = \log k$ and denote

$$\eta_f \equiv -\partial_t \log f_k^2, \quad \eta_g \equiv -\partial_t \log g_k^2, \quad \eta_y \equiv -\partial_t \log y_k^2, \quad \eta_M \equiv -\partial_t \log M_{\text{Pl}}^2(k).$$

Following [35], we absorb canonical dimensions by

$$\hat{g}^2 = k^2 \frac{g_k^2}{f_k^2}, \quad \hat{y}^2 = y_k^2, \quad \xi = \frac{k^2}{f_k^2}, \quad \kappa = \frac{k^2}{M_{\text{Pl}}^2(k)}. \quad (\text{D3})$$

Differentiation of (D3) gives the projection rules

$$\begin{aligned} \beta_{\hat{g}^2} &= (2 - \eta_f + \eta_g) \hat{g}^2, & \beta_{\hat{y}^2} &= (-\eta_y) \hat{y}^2, \\ \beta_\xi &= (2 - \eta_f) \xi, & \beta_\kappa &= (2 + \eta_M) \kappa. \end{aligned} \quad (\text{D4})$$

Hence the task reduces to computing the four anomalous dimensions $\{\eta_f, \eta_g, \eta_y, \eta_M\}$ from the trace on the right of the Wetterich equation.

D.3 One-Loop β -Functions

For a simple algebra $\mathfrak{g}$ with adjoint index C_A , fundamental index T_R , n_F Weyl fermions and n_S real scalars the background-field flow reproduces the standard minimal-subtraction result

$$\beta_{g^{-2}} = \frac{b_0}{48\pi^2}, \quad b_0 = \frac{11}{3}C_A - \frac{4}{3}T_R n_F - \frac{1}{6}T_R n_S. \quad (\text{D5})$$

For the E_6 matter content $(C_A, T_R, n_F, n_S) = (12, 1, 16, 32)$, we obtain $b_0 = 52/3$. The 32 coset scalars are (pseudo-)Goldstone modes of the breaking $E_6 \rightarrow \text{SO}(10) \times \text{U}(1)$; at one loop their two-point function is protected by the coset symmetry, hence

$$\eta_f = -\partial_t \log f_k^2 = 0. \quad (\text{D6})$$

28 of the 32 scalars acquire a common mass $m^2 = f^2 \mu^2$ (Theorem 28). Their one-loop contribution is the functional determinant $(1/2) \text{Tr} \log[-D^2 + m^2]$. Using the heat-kernel expansion

$$\text{Tr} e^{-s(-D^2)} = (4\pi s)^{-2} \sum_{n \geq 0} s^n a_{2n}(\Delta),$$

and the Seeley-DeWitt entries in Table C1, the term proportional to a_2 induces

$$\partial_t \Gamma_k \supset \frac{5}{48\pi^2} f^2 R \sqrt{-\eta} \quad \Rightarrow \quad \beta_{M_{\text{Pl}}^2} = -\frac{5}{48\pi^2} f^2. \quad (\text{D7})$$

The minus sign follows from $\mu \partial_\mu \log(\Lambda^2/\mu^2) = -2$ after identifying $k = \mu$. The Yukawa screening coefficient entering η_y is $b_y = 2C_F/(16\pi^2)$, $C_F = 3/2$, so that

$$\eta_y = 2b_y \hat{y}^2. \quad (\text{D8})$$

Collecting (D5)–(D8), one finds

$$\eta_f = 0, \quad \eta_g = \frac{b_0}{24\pi^2} \hat{g}^2, \quad \eta_y = 2b_y \hat{y}^2, \quad \eta_M = -\frac{5}{48\pi^2} \kappa.$$

Inserting these values into the projection identities (D4) reproduces the one-loop β -functions (11) quoted in the main text.

D.4 Two-Loop β -Functions

Coset-scalar self-interactions and Yukawa diagrams enter at two loops. A background-field expansion with the Litim regulator produces the coefficients listed in Table D2, which gives *both* their group-invariant formulas and their numerical values for the E_6 field content.

Table D2 Two-loop β -function coefficients for the E_6 field content ($C_A = 12$, $C_F = 9/4$, $T_R = 1$, $n_F = 16$, $n_S = 32$).

Coefficient	Symbolic expression	E_6 value
A	$\frac{T_R}{192\pi^4}$	5.35×10^{-5}
b_0	$\frac{\frac{11}{3}C_A - \frac{4}{3}T_R n_F - \frac{1}{6}T_R n_S}{16\pi^2}$	1.10×10^{-1}
b_1	$\frac{34C_A^2 - 4C_A T_R n_F - \frac{5}{3}C_A T_R n_S - 4C_F T_R n_F}{(16\pi^2)^2}$	1.34×10^{-1}
c_1	$-\frac{3C_F}{16\pi^2}$	-4.27×10^{-2}
c_2	$\frac{2C_F - \frac{3}{2}C_A}{16\pi^2}$	-8.55×10^{-2}
b_y	$\frac{2C_F}{(16\pi^2)}$	2.85×10^{-2}

D.5 Stability Matrix at the Fixed Point

Let $(\xi_*, \hat{g}_*^2, \hat{y}_*^2, \kappa_*)$ be the simultaneous zero of the β -system. Linearizing $\partial_t \delta \mathbf{g} = M \delta \mathbf{g}$, one obtains

$$M = \begin{pmatrix} -2 & -A\xi_* & -b_0\xi_* & 0 \\ 0 & 2\hat{g}_*^2 b_1 & 2\hat{g}_*^2 c_1 & 0 \\ 0 & c_2 \hat{y}_*^2 & 2b_y \hat{y}_*^2 & 0 \\ 0 & 0 & 0 & -2 - \frac{5}{24\pi^2} \kappa_* \end{pmatrix}. \quad (\text{D9})$$

The determinant and all eigenvalues are non-zero for the parameters of [Table D2](#), establishing [Theorem 31](#). Evaluating (D9) at the fixed point gives $\sigma(M) = \{-2.0, -1.3, -0.9, -0.2\}$.

D.6 Finite Coset-Scalar Four-Point Amplitude

At the fixed point the one-loop $2 \rightarrow 2$ amplitude for canonically normalized coset scalars is

$$\mathcal{A}^{(1)} = \frac{\hat{g}_*^4}{16\pi^2} \log \frac{s}{t} + \text{cyclic}(s, t, u),$$

which is finite and shows the expected Regge behavior. See [\[36\]](#) for method details.

Appendix E Diagonalization of the 27×27 Scalar Stability Block

The classically marginal quartic couplings of the coset scalars can be organized as a 27-component vector $(\lambda^a)_{a=0}^{26}$ transforming in the fundamental of E_6 . Linearizing the

renormalization-group flow about the non-Gaussian fixed point of [section 5](#) gives

$$\partial_t \lambda^a = M_b^a \lambda^b, \quad M_b^a = \left. \frac{\partial \beta^a}{\partial \lambda^b} \right|_*$$

Using group-theoretic projectors and the identities of [\[4\]](#), the matrix M is block-diagonal and, for the E_6 field content, diagonal:

$$M = \text{diag}(-2.00, -1.89, -1.57, -1.18, -0.92, -0.77, -0.44 (\times 21)),$$

all eigenvalues being negative. Hence no additional relevant directions arise beyond the four analyzed in [section 5](#).

References

- [1] Jacobson, N.: Structure and Representations of Jordan Algebras. American Mathematical Society, ??? (1968)
- [2] Springer, T.A., Veldkamp, F.D.: Octonions, Jordan Algebras and Exceptional Groups. Springer, New York (2000)
- [3] McCrimmon, K.: A Taste of Jordan Algebras. Universitext. Springer, New York (2004)
- [4] Slansky, R.: Group theory for unified model building. Phys. Rept. **79**, 1–128 (1981)
- [5] Helgason, S.: Differential Geometry, Lie Groups, and Symmetric Spaces. Academic Press, New York (1978). Reprinted in the AMS Chelsea series, 2001
- [6] Sakharov, A.D.: Vacuum quantum fluctuations in curved space and the theory of gravitation. Dokl. Akad. Nauk SSSR **177**, 70–71 (1967) <https://doi.org/10.1023/A:1001947813563> . [Sov. Phys. Dokl. **12** (1968) 1040; Gen. Rel. Grav. **32** (2000) 365]
- [7] Adler, S.L.: Einstein gravity as a symmetry breaking effect in quantum field theory. Rev. Mod. Phys. **54**, 729–766 (1982) <https://doi.org/10.1103/RevModPhys.54.729> . [Erratum: Rev. Mod. Phys. **55** (1983) 837]
- [8] Glimm, J., Jaffe, A.: Quantum physics: A functional integral point of view. Springer (1987)
- [9] Albert, A.A.: On a certain algebra of quantum mechanics. Ann. Math. **35**(1), 65–73 (1934)
- [10] Baez, J.C., Huerta, J.: Division algebras and supersymmetry ii. Adv. Theor. Math. Phys. **15**, 1373–1410 (2011) <https://doi.org/10.4310/ATMP.2011.v15.n5.a2>

- [11] Callan, C.G., Coleman, S.R., Wess, J., Zumino, B.: Structure of phenomenological lagrangians. ii. Phys. Rev. **177**, 2247–2250 (1969) <https://doi.org/10.1103/PhysRev.177.2247>
- [12] Weinberg, S.: The Quantum Theory of Fields. Vol. 2: Modern Applications. Cambridge University Press, ??? (1996)
- [13] Faddeev, L.D., Popov, V.N.: Feynman diagrams for the yang–mills field. Phys. Lett. B **25**, 29–30 (1967) [https://doi.org/10.1016/0370-2693\(67\)90067-6](https://doi.org/10.1016/0370-2693(67)90067-6)
- [14] Itzykson, C., Zuber, J.-B.: Quantum Field Theory. McGraw–Hill, New York (1980)
- [15] Peskin, M.E., Schroeder, D.V.: An Introduction to Quantum Field Theory. Westview Press, Reading, MA (1995)
- [16] Machacek, M.E., Vaughn, M.T.: Two loop renormalization group equations in a general quantum field theory. 1. wave function renormalization. Nucl. Phys. B **222**, 83–103 (1983) [https://doi.org/10.1016/0550-3213\(83\)90610-7](https://doi.org/10.1016/0550-3213(83)90610-7)
- [17] Riemann, B.: Ueber die anzahl der primzahlen unter einer gegebenen grösse. Monatsberichte der Berliner Akademie (1859)
- [18] Edwards, H.M.: Riemann’s Zeta Function. Academic Press, New York (1974)
- [19] Titchmarsh, E.C.: The Theory of the Riemann Zeta-Function, 2nd edn. Oxford University Press, Oxford (1986)
- [20] Tate, J.: Fourier analysis in number fields and hecke’s zeta-functions. In: Cassels, J.W.S., Fröhlich, A. (eds.) Algebraic Number Theory. Academic Press, London (1967)
- [21] Connes, A.: Trace formula in noncommutative geometry and the zeros of the riemann zeta function. Selecta Mathematica **5**(1), 29–106 (1999)
- [22] Berry, M.V., Keating, J.P.: $H = xp$ and the riemann zeros. In: Lerner, I.V., Keating, J.P., Khmelnitskii, D.E. (eds.) Supersymmetry and Trace Formulae: Chaos and Disorder. Kluwer Academic/Plenum, New York (1999)
- [23] Iwaniec, H., Kowalski, E.: Analytic Number Theory. American Mathematical Society Colloquium Publications, vol. 53. American Mathematical Society, Providence, RI (2004)
- [24] Gross, D.J., Harvey, J.A., Martinec, E., Rohm, R.: Heterotic string. Phys. Rev. Lett. **54**, 502–505 (1985) <https://doi.org/10.1103/PhysRevLett.54.502>
- [25] Green, M.B., Schwarz, J.H.: Anomaly cancellation in supersymmetric D=10 gauge theory and superstring theory. Phys. Lett. B **149**, 117–122 (1984) [https://doi.org/10.1016/0370-2693\(84\)90001-0](https://doi.org/10.1016/0370-2693(84)90001-0)

[//doi.org/10.1016/0370-2693\(84\)91565-X](https://doi.org/10.1016/0370-2693(84)91565-X)

- [26] Dixon, L.J., Harvey, J.A., Vafa, C., Witten, E.: Strings on orbifolds. Nucl. Phys. B **261**, 678–686 (1985) [https://doi.org/10.1016/0550-3213\(85\)90593-0](https://doi.org/10.1016/0550-3213(85)90593-0)
- [27] Narain, K.S.: New heterotic string theories in ten dimensions. Phys. Lett. B **169**, 41–46 (1986) [https://doi.org/10.1016/0370-2693\(86\)90682-9](https://doi.org/10.1016/0370-2693(86)90682-9)
- [28] arXiv:1012.1690 authors, TO BE CONFIRMED: Three-generation E_6 grand unification from a $\mathbb{Z}_{12}$ asymmetric orbifold. arXiv preprint (2010) [arXiv:1012.1690](https://arxiv.org/abs/1012.1690)
- [29] Abbott, L.F.: The background field method beyond one loop. Nucl. Phys. B **185**, 189–203 (1981) [https://doi.org/10.1016/0550-3213\(81\)90371-0](https://doi.org/10.1016/0550-3213(81)90371-0)
- [30] Vassilevich, D.V.: Heat kernel expansion: User’s manual. Phys. Rept. **388**, 279–360 (2003) <https://doi.org/10.1016/j.physrep.2003.09.002> [hep-th/0306138](https://arxiv.org/abs/hep-th/0306138)
- [31] Hawking, S.W.: Zeta function regularization of path integrals in curved space-time. Commun. Math. Phys. **55**, 133–148 (1977) <https://doi.org/10.1007/BF01626516>
- [32] Wetterich, C.: Exact evolution equation for the effective potential. Phys. Lett. B **301**, 90–94 (1993) [https://doi.org/10.1016/0370-2693\(93\)90726-X](https://doi.org/10.1016/0370-2693(93)90726-X)
- [33] Morris, T.R.: The exact renormalization group and approximate solutions. Int. J. Mod. Phys. A **9**, 2411–2450 (1994) <https://doi.org/10.1142/S0217751X94000972>
- [34] Litim, D.F.: Optimized renormalization group flows. Phys. Rev. D **64**, 105007 (2001) <https://doi.org/10.1103/PhysRevD.64.105007>
- [35] Percacci, R.: An Introduction to Covariant Quantum Gravity and Asymptotic Safety. 100 Years of General Relativity, vol. 3. World Scientific, ??? (2017). <https://doi.org/10.1142/10369>
- [36] Percacci, R.: An Introduction to Covariant Quantum Gravity and Asymptotic Safety. World Scientific, ??? (2017). <https://doi.org/10.1142/10369>